

CO₂-dependent opening of Connexin 43 hemichannels

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This **important** study reveals that connexin43 (Cx43) hemichannels are directly activated by CO₂ through a conserved carbamylation motif, extending a mechanism previously described for β-connexins to α-connexins. The evidence is **convincing**, supported by complementary biochemical and electrophysiological analyses showing CO₂-induced hemichannel opening and ATP release in cultured cells and hippocampal slices. These findings advance our understanding of connexin regulation by metabolic gases and will be of broad interest to researchers studying cell communication, neural signaling, and gasotransmitter biology.

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Abstract

Sequence and structure comparisons between alpha and beta connexins, Cx43 and Cx26, revealed that Cx43 has a motif, the carbamylation motif, that confers CO₂-sensitivity on a subset of beta connexins. By using a fluorescent dye loading assay, whole cell patch clamp recordings and real time measurement of ATP release via GRAB_{ATP} we have demonstrated that Cx43 hemichannels open in a highly CO₂ sensitive manner over the range 20 to 70 mmHg. Mutational analysis confirms that the equivalent residues to those in Cx26 that have been shown to be involved in mediating the effects of CO₂ on gating of hemichannels and gap junction channels, also mediate Cx43 hemichannel gating. Our data predicts that Cx43 will be partially open and able to release ATP at resting physiological levels of PCO₂. We have tested this prediction in acute hippocampal slices, by showing that CO₂-dependent enhancement of synaptic transmission can be blocked by the Cx43-selective mimetic peptide Gap26. Our data resolves an inconsistency in the literature between *in vivo* studies suggesting that Cx43 hemichannels are at least partially open at rest, and *in vitro* studies, performed in the absence of HCO₃⁻/CO₂ buffering that show Cx43 hemichannels are shut. Our evidence suggests that the ancestral gene that duplicated to give the alpha and beta connexin clades must have possessed the carbamylation motif. CO₂ sensitivity is thus a fundamental ancient characteristic of several connexins that has been lost in more recently derived members of this gene family.

Introduction

There are 21 connexin genes encoded in the human genome. Connexins are transmembrane proteins that can form hemichannels and gap junction channels. The existence of 21 isoforms suggests functional specialisation across the gene family (Sohl & Willecke, 2004 [\[1\]](#)). On the basis of molecular phylogeny, the connexins have been divided into 4 broad clades - the alpha, beta, and gamma and delta groupings (Mikalsen *et al.*, 2021 [\[2\]](#)).

Certain members of the β -clade, Cx26, Cx30 and Cx32, are directly sensitive to CO₂ (Huckstepp *et al.*, 2010 [\[3\]](#); Dospinescu *et al.*, 2019 [\[4\]](#)). The mechanism of CO₂ sensitivity has best been studied in Cx26. While Cx26 hemichannels are opened in response to CO₂ (Huckstepp *et al.*, 2010 [\[3\]](#)), CO₂ closes Cx26 gap junction channels (Nijjar *et al.*, 2021 [\[5\]](#)). The CO₂-dependent gating of Cx26 hemichannels and gap junction channels involves carbamylation of a specific lysine, K125, which is part of a critical carbamylation motif, ₁₂₅KVRIEGS₁₃₁ (Meigh *et al.*, 2013 [\[6\]](#)). This motif is essential for the CO₂-sensitivity of Cx26, as it allows the formation of a salt bridge between the carbamylated K125 and R104 of the neighbouring subunit (Meigh *et al.*, 2013 [\[6\]](#)). The formation of this “carbamate” bridge induces conformational changes, ultimately leading to the opening of the hemichannel or closing of the gap junction channel (Meigh *et al.*, 2013 [\[6\]](#); Nijjar *et al.*, 2021 [\[5\]](#); Brotherton *et al.*, 2022 [\[7\]](#); Brotherton *et al.*, 2024 [\[8\]](#)). Mutation of K125 to arginine, an amino acid that cannot be carbamylated, or R104 to alanine so that it cannot form a carbamate bridge, removes the ability of CO₂ to open Cx26 hemichannels (Meigh *et al.*, 2013 [\[6\]](#)) or close Cx26 gap junction channels (Nijjar *et al.*, 2021 [\[5\]](#); Nijjar *et al.*, 2025 [\[9\]](#)). Furthermore, the introduction of the carbamylation motif into a CO₂-insensitive connexin, Cx31, results in CO₂-dependent hemichannel opening of Cx31 (Meigh *et al.*, 2013 [\[6\]](#)). Other members of the β -clade, Cx30 and Cx32, are also CO₂-sensitive, but have different sensitivity profiles (Huckstepp *et al.*, 2010 [\[3\]](#); Dospinescu *et al.*, 2019 [\[4\]](#)). Evolutionary comparisons show that the carbamylation motif has been conserved over 400 million years (Dospinescu *et al.*, 2019 [\[4\]](#)) and appears to be a necessary and sufficient feature for the CO₂-sensitivity of connexins.

Sequence comparisons of the β -connexins and α -connexins, revealed the presence of a potential carbamylation motif within the α -connexin, Cx43. Cx43 has a motif (₁₄₄KVKMRGG₁₅₀) homologous to the motif in Cx26 (₁₂₅KVRIEGS₁₃₁) suggesting that it might also be sensitive to CO₂. In this study we show that Cx43 hemichannels are indeed directly sensitive to changes in the partial pressure of CO₂ (PCO₂) over the physiological concentration range. Unlike Cx26, the CO₂-dependent opening mechanism of Cx43 is more complex, involving additional residues beyond the homologs of K125 and R104 in Cx26. Our discovery that Cx43 hemichannels are directly CO₂ sensitive has major physiological implications and may explain why Cx43 hemichannels appear to be open under physiological conditions but closed *in vitro* where they are usually studied in the absence of CO₂/HCO₃ buffers.

Results

Identification of a potential carbamylation motif in Cx43

Cx43, an alpha-connexin, possesses a carbamylation motif homologous to that of the beta-connexins (Fig 1A [\[10\]](#)). AlphaFold3 (AF3) predicts a structure very close to the experimentally determined structures for Cx43 (Lee *et al.*, 2023a [\[11\]](#); Qi *et al.*, 2023 [\[12\]](#)) but has the advantage of including portions of the structure that have not been experimentally resolved. This is a valid approach, as the *de novo* predictions by AF3 for the cytoplasmic loop of Cx26 have been supported by a recent experimentally-derived structure (Brotherton *et al.*, 2024 [\[8\]](#)). By performing

alignments of the Cx43 and Cx26 AF3 structures (**Fig 1B**), we found that two residues in Cx43, K144 and K105, aligned with the residues in Cx26 that are essential for hemichannel opening to CO₂, K125 and R104 (**Fig 1C**). The predicted 3D orientation of the alpha-helices involved are positioned in a similar manner to allow potential interaction between K144 and K105 following carbamylation. Cx43 also presents another feature that is similar to Cx26: the high incidence of Lys residues. In Cx26, 5 lysine residues in the cytoplasmic loop can be carbamylated (Nijjar *et al.*, 2025), suggesting an environment favourable for this to happen. The presence of multiple Lys residues in the cytoplasmic loop of Cx43 implies that the local environment may be conserved to facilitate the required pKa for CO₂ binding and carbamate bridge formation.

Cx43 hemichannels are opened by CO₂

We used an established dye-loading protocol, previously used for CO₂ connexin studies (Huckstepp *et al.*, 2010; Meigh *et al.*, 2013; Meigh *et al.*, 2014; de Wolf *et al.*, 2016; de Wolf *et al.*, 2017; Cook *et al.*, 2019; Dospinescu *et al.*, 2019; van de Wiel *et al.*, 2020), to test whether Cx43 hemichannels are CO₂ sensitive. We observed significant dye-loading in Cx43-expressing cells when exposed to a high CO₂ stimulus (70 mmHg) when starting from a baseline of 20 mmHg (MW test, PCO₂ 20 vs 70 mmHg: $p = 0.008$, $n=5$, **Fig 2A**). We used a zero Ca²⁺ aCSF to unblock hemichannels as a positive control to demonstrate that for WT Cx43, the zero Ca²⁺ stimulus and 70 mmHg PCO₂ gave a similar amount of dye loading (MW test: p -value = 0.726, $n=5$).

To further support these findings, we next used whole cell patch clamp recordings from HeLa cells expressing Cx43 (**Fig 2B**). We set the holding potential to -50 mV and gave regular steps to -40 mV to measure whole cell conductance. From a baseline of 35 mmHg PCO₂, an increase in an outward conductance was observed when switching to 55 or 70 mmHg aCSF (**Fig 2B**). On switching to 20 mmHg aCSF, a small conductance decrease and reduction of outward current was observed (**Fig 2B**). Untransfected parental HeLa cells did not display changes in outward current to changes in PCO₂. However, some parental HeLa cells exhibited a slow increase in an inward current on transfer from 35 to 70 mmHg (**Fig 2B**). This was quite distinct from the outward current observed in Cx43 expressing cells (**Fig 2B**). Overall, the patch clamp data predicted a dose response that could be fitted by the Hill equation with a Hill coefficient of 6 and an EC₅₀ of 43 mmHg. This is similar to Cx26 hemichannels which also display a steep CO₂ dose response curve with a Hill coefficient of >4 (Huckstepp *et al.*, 2010; de Wolf *et al.*, 2017; van de Wiel *et al.*, 2020).

To assess whether the CO₂-dependent opening of Cx43 resulted in ATP release, we co-transfected HeLa cells with the genetically encoded ATP sensor - GRAB_{ATP} (Wu *et al.*, 2022). CO₂ dependent increase in ATP release was indeed observed ($n=22$, **Fig 2C**). We observed an increase in GRAB_{ATP} fluorescence on going from 20 to 35 mmHg PCO₂ and further to 55 mmHg PCO₂ (**Fig 2C**). While ATP release at 70 mmHg was slightly less than that at 55 mmHg, it remained markedly higher than at 35 mmHg PCO₂ (**Fig 2C**). By contrast, HeLa cells that were not transfected with Cx43 showed no responses to any applied CO₂ concentration as concluded from GRAB_{ATP} experiments ($n=14$, **Fig 2D**). Overall, our results show through three independent assays that Cx43 hemichannels are opened by modest changes in PCO₂ around the physiological norm and will be partially open at a PCO₂ of 35 mmHg (**Fig 2**). Once opened by CO₂, Cx43 hemichannels display permeability to ATP (**Fig 2C**).

Cx43 displays sensitivity to intracellular pH via a sensor located in the C-terminus (Ek-Vitorin *et al.*, 1996). We assessed the effect of truncations of the C-terminus of Cx43 (truncated after residue 256, Cx43¹⁻²⁵⁶) that remove this pH sensor. Both CO₂ and high K⁺ induced ATP release from HeLa cells that expressed Cx43¹⁻²⁵⁶ (**Fig 2, figure supplement 1**). We also measured the effect of changing PCO₂ on intracellular pH (pH_i) of HeLa cells. We found that transferring from 35 to 55 or

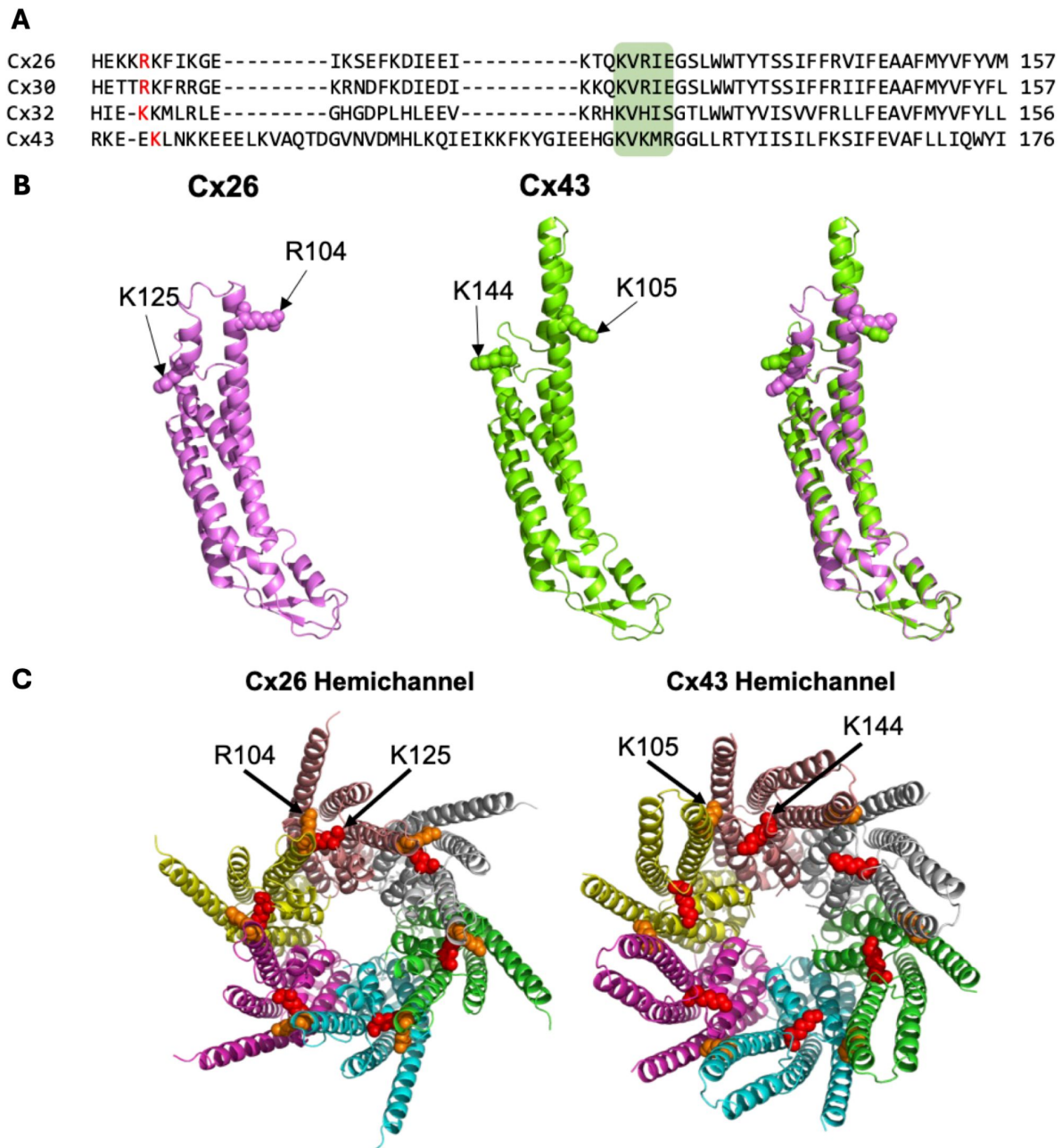


Figure 1

Structures of Cx43 and Cx26 predicted by AF3.

(A) α - and β -connexin sequence alignment with the known CO₂-sensitive connexins at the top, Cx26, Cx30 and Cx32, followed by the α -connexin Cx43. The carbamylation motif is highlighted by the green box. (B) Cx26 and Cx43 AlphaFold3 structural alignments. (C) Cx26 and Cx43 hemichannel structural prediction.

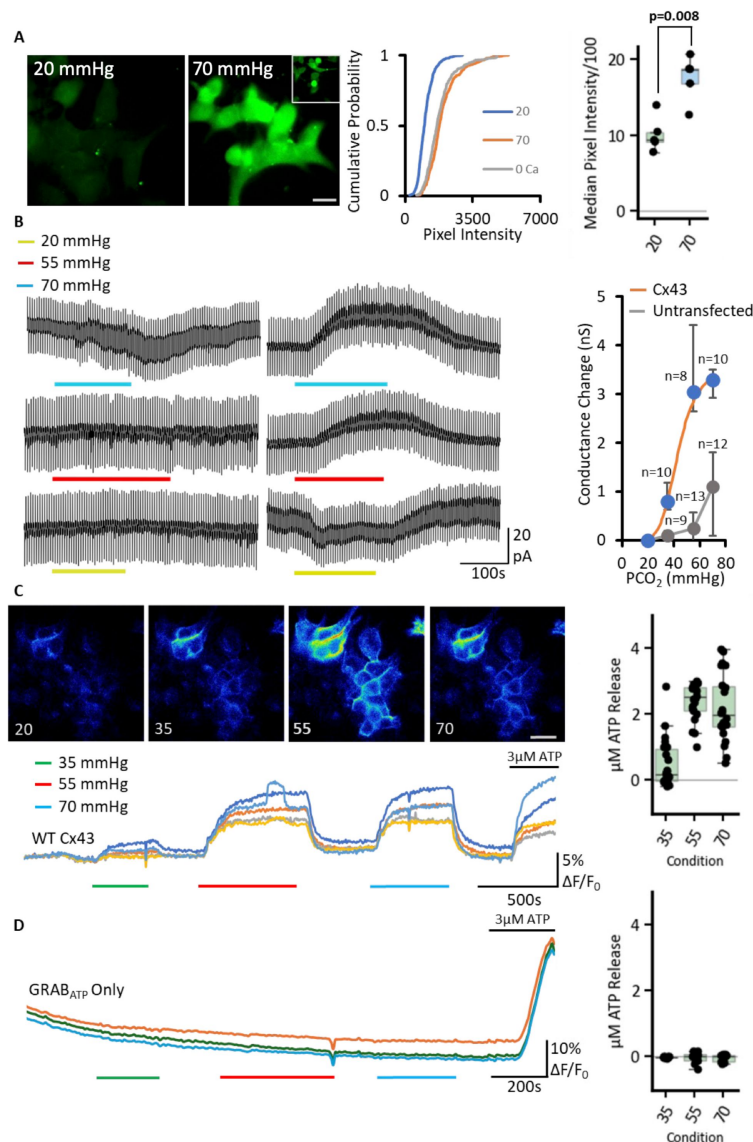


Figure 2

Cx43 hemichannels can be opened by an increase in PCO₂.

The CO₂ sensitivity of Cx43 was assayed using three different methods. **(A)** Dye-loading with CBF from 20 (left) to 70 mmHg PCO₂ (right). The inset represents the 0 Ca²⁺ control, white scale bar is 20 μm. Cumulative probability graph represents all the data points measured with more than 200 cells per condition, box plot shows the medians from each independent transfection. **(B)** Whole cell patch recordings performed on untransfected parental HeLa cells and HeLa cells expressing Cx43 at a holding potential of -50 mV with steps to -40 mV to measure whole cell conductance. The baseline was set to 35 mmHg PCO₂ and three different concentrations were used (20, 55, 70 mmHg). The absolute conductance change evoked by a change in PCO₂ was measured and plotted as a dose response curve (right side, median with interquartile range) with 20 mmHg being assigned to zero and the absolute value of all conductance changes plotted relative to this (see Methods for details). The points were fitted by the Hill equation H=6, EC₅₀=43 mmHg (blue points, orange curve). The data for untransfected parental HeLa cells was plotted the same way (grey). **(C)** The genetically-encoded ATP sensor GRAB_{ATP} was co-transfected alongside Cx43 into HeLa cells (n=22). Representative images at different PCO₂ concentrations of the cells are shown, grey scale bar is 20 μm. Traces from the recording the images were selected from are shown; each line represent a different cell, the fluorescence was normalised by dividing all values by the baseline median pixel intensity before plotting. Box plot shows the μM ATP release as determined by normalisation to the 3 μM control solution applied. **(D)** Parental HeLa cells transfected only with GRAB_{ATP} do not exhibit CO₂ dependent ATP release (n=14), summary box plot shows the μM ATP release as determined by normalisation to 3 μM ATP calibration.

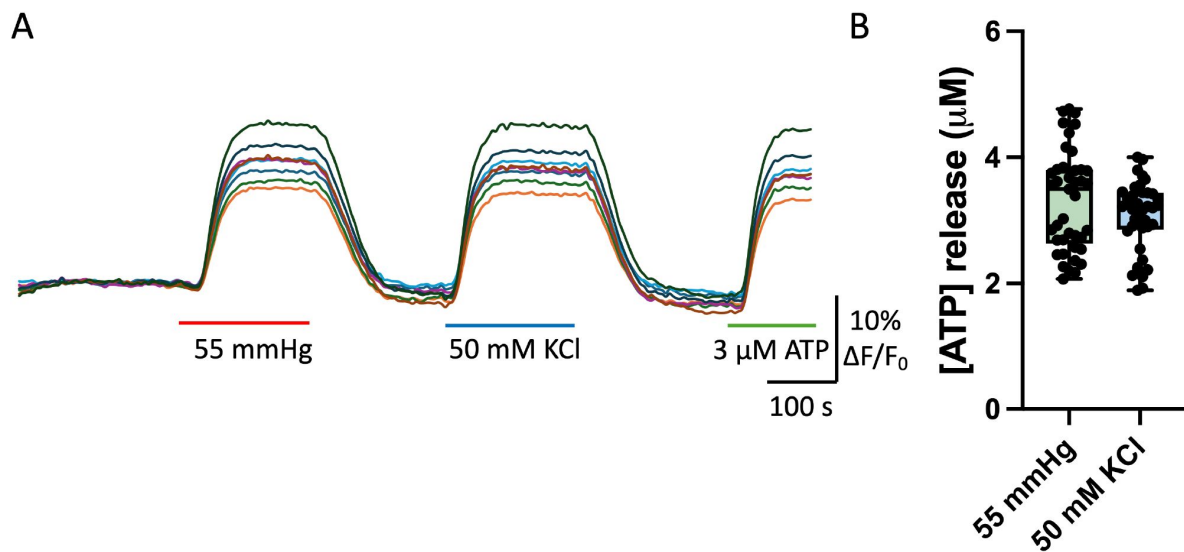


Figure 2, figure supplement 1

Cx43 hemichannels with a truncated C-terminus (Cx43¹⁻²⁵⁶) retain their sensitivity to CO₂ and depolarisation.

(A) Changes in GRAB_{ATP} fluorescence induced by changing PCO₂ from 20 to 55 mmHg or KCl from 3 to 50 mM. **(B)** Summary graph showing the concentration of ATP release in response to 55 mmHg PCO₂ and 50 mM KCl, n=41 cells from 3 independent transfections.

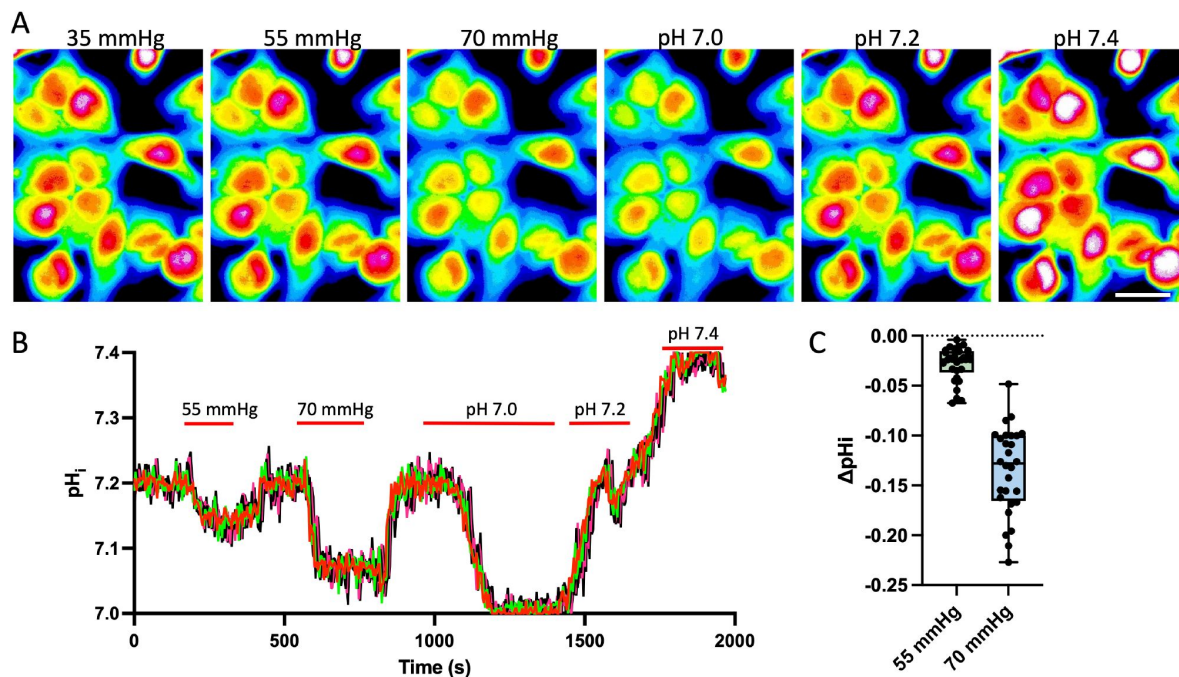


Figure 2, figure supplement 2

Effect of hypercapnic solutions on intracellular pH (pH_i) of parental HeLa cells measured by BCECF fluorescence.

(A) Images showing BCECF fluorescence at 35, 55 and 70 mmHg, and then calibration to pH 7.0, 7.2 and 7.4 of the same cells after treatment with nigericin. (B) Quantification of the changes in fluorescence. C) Summary plot of the change in pH_i induced by 55 and 70 mmHg PCO₂. 3 independent seeds of cells, n=30.

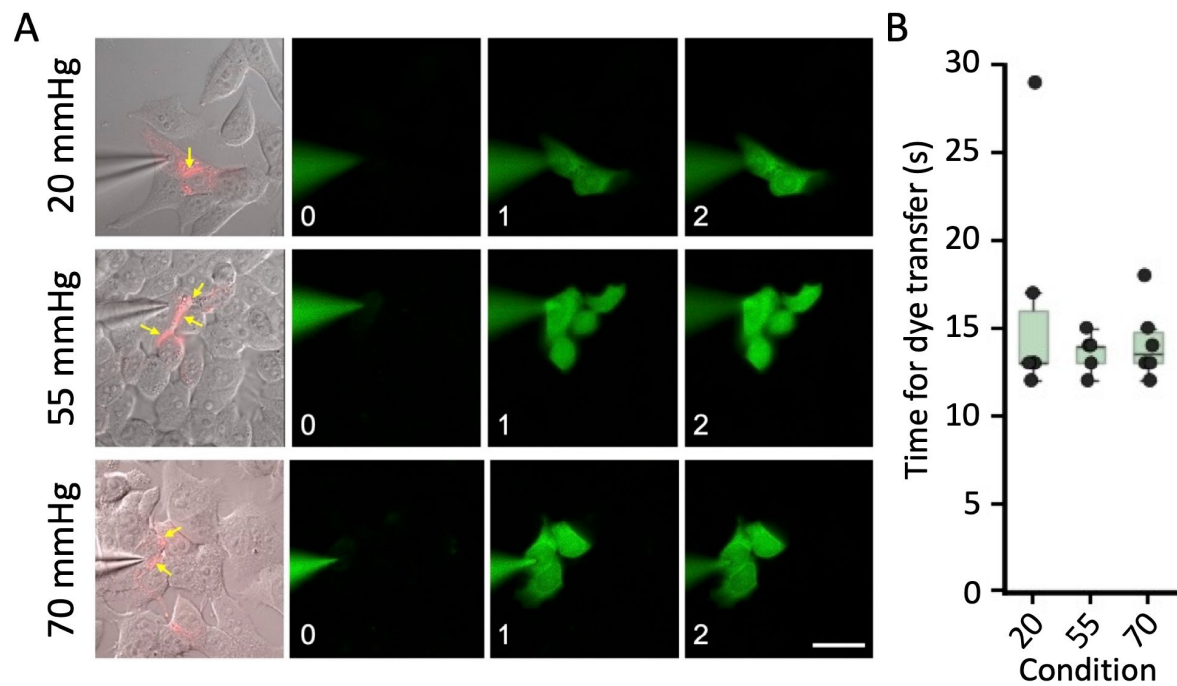


Figure 2, figure supplement 3

Cx43 gap junction channels are insensitive to changes in PCO₂.

(A) DIC image with mCherry fluorescence superimposed. Gap junctions are evident as red stripes between cells (yellow arrows). The fluorescence images show the patch pipette filled with NBDG its loading into the recorded cell and the neighbouring cells coupled via gap junctions. The numbers in each bottom left corner are the time of the recording from whole cell breakthrough in minutes. Scale bar 20 μ m. **(B)** Summary graphs showing the time for the acceptor cell to reach 10% of the fluorescence of the donor cell at three different levels of PCO₂ (n=6 for each condition).

70 mmHg respectively induced median changes in pH_i of -0.02 (95% CI $-0.02, -0.03$) and -0.13 (CI $-0.10, -0.15$; **Fig 2, figure supplement 2**). These results show that the effects of PCO_2 on Cx43 gating are independent of those involved in its pH sensitivity.

Gap Junction Channels (GJCs) of Cx26 are closed by CO_2 also acting via the carbamylation motif. We therefore tested whether Cx43 GJCs might also be CO_2 sensitive by using a dye transfer assay (Nijjar *et al.*, 2021). Cx43 GJCs allowed equally fast permeation of dye at all levels of PCO_2 and therefore, unlike the hemichannel, were insensitive to CO_2 (**Fig 2, figure supplement 3**).

Identifying the residues required for CO_2 -sensitivity of Cx43

Having established the dose-response characteristics of Cx43 to CO_2 , we next examined the potential underlying mechanisms for CO_2 -dependent gating via targeted mutagenesis of possible key residues, starting with the equivalents of those known to be involved in the CO_2 -sensitivity of Cx26 (**Fig 1**).

Mutation of K144 to glutamine (K144Q), an amino acid that cannot be carbamylated, partially reduced the CO_2 -sensitivity of the channel: the 70 mmHg stimulus resulted in significantly less increase in dye loading than the zero Ca^{2+} positive control (**Fig 3A**, MW test: $p = 0.048$, $n=5$). However, mutation of Lys105 to a non-charged residue (K105Q) resulted in hemichannels that gave a similar increase in dye loading to the zero Ca^{2+} positive control (**Fig 3B**, MW test: $p = 0.111$, $n=5$). In Cx26, mutation of the residues involved in carbamylation (K125 and R104) results in complete loss of CO_2 sensitivity (Meigh *et al.*, 2013; Nijjar *et al.*, 2021). The lack of such a complete effect on the CO_2 sensitivity of Cx43 implies that the mechanism is more complex and may involve further residues. Interestingly, mutation of Lys144 to Arg (K144R) completely abolished the CO_2 sensitivity of Cx43 hemichannels as measured by dye loading (**Fig 3, figure supplement 1**). The difference between the effect of mutating Lys144 to Arg and Gln on CO_2 sensitivity may arise because the substitution to Arg introduces a positive charge. Since K144 appears to be carbamylated, its primary amine must remain unprotonated so that the unpaired electrons of the nitrogen can mount a nucleophilic attack on the carbon of CO_2 to form the carbamate bond. Thus, the K144Q mutation may preserve this charge and be a better substitute that removes the capacity for carbamylation while preserving the overall charge within this region of the protein. Consequently, we have predominantly used the Lys to Gln substitution in this study.

We further inspected the AF3 structure prediction and identified two further lysine residues of interest (**Fig 3E**). K109 in Cx43 aligns with K108 of Cx26, a residue known to be carbamylated and required for Cx26 gap junction closure (Nijjar *et al.*, 2025). AF3 also suggests that K109 could potentially interact with K234 following any carbamylation (**Fig 3E**). We therefore mutated both residues to Gln to test their possible involvement in CO_2 sensitivity. The dye-loading assay showed that K109Q remained CO_2 sensitive (**Fig 3C**, MW test vs zero Ca^{2+} control, $p = 0.274$, $n=5$). However, the mutation K234Q displayed impaired CO_2 sensitivity (**Fig 3D**, MW test vs zero Ca^{2+} control: $p = 0.016$, $n=5$).

As the dye loading assay suggested that single mutations were on the whole rather ineffective in abolishing CO_2 sensitivity, we assayed ATP release via GRAB_{ATP} to gain further supporting evidence. We found that CO_2 -evoked ATP release remained dose dependent and very similar to the WT Cx43 in the K105Q ($n=23$) and K144Q ($n=12$) mutants (**Fig 4A, C**, **Figure 4, figure supplement 1**). Cx43 hemichannels with the K109Q mutation appeared to have their sensitivity to CO_2 shifted to lower values of PCO_2 ($n=20$), appearing to give maximal ATP release at 35 mmHg (**Fig 4B, E**, **Fig 4 figure supplement 1**). K234Q Cx43 hemichannels had greatly reduced sensitivity to CO_2 ($n=15$, **Fig 4D, F**). To check that K234Q did not have some non-specific effect on

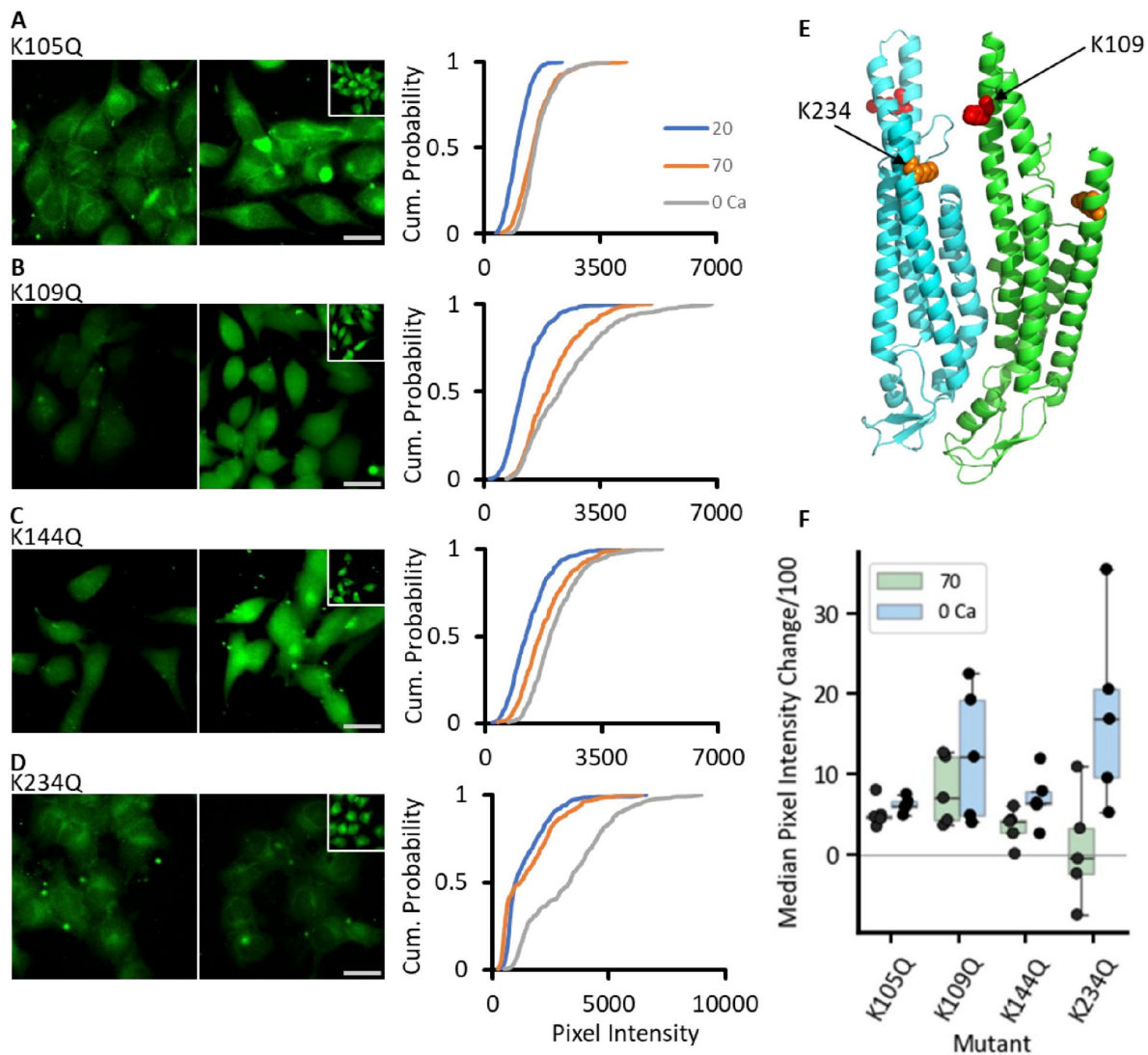


Figure 3

The effect of single Lys mutations on the CO₂ sensitivity of Cx43.

HeLa cells transfected with the mutant variant of Cx43 were subjected to the dye-loading protocol. Images of dye loading in response to CO₂ and the 0 Ca²⁺ positive control (insets), together with the cumulative probability plots are shown for K105Q (A), K109Q (B), K144Q (C), K234Q (D). (E) AlphaFold3 prediction of Cx43 with hypothesised carbamylation bridge residues highlighted – K234 in orange and K109 in red. (F) Summary box plots for dye-loading showing the change in median pixel intensity from 20 mmHg PCO₂ for each condition (70 mmHg PCO₂ – green and 0 Ca²⁺ blue, n=5) for WT (recalculated data from Fig 2A for comparison) and each mutation. Scale bars are 20 μm.

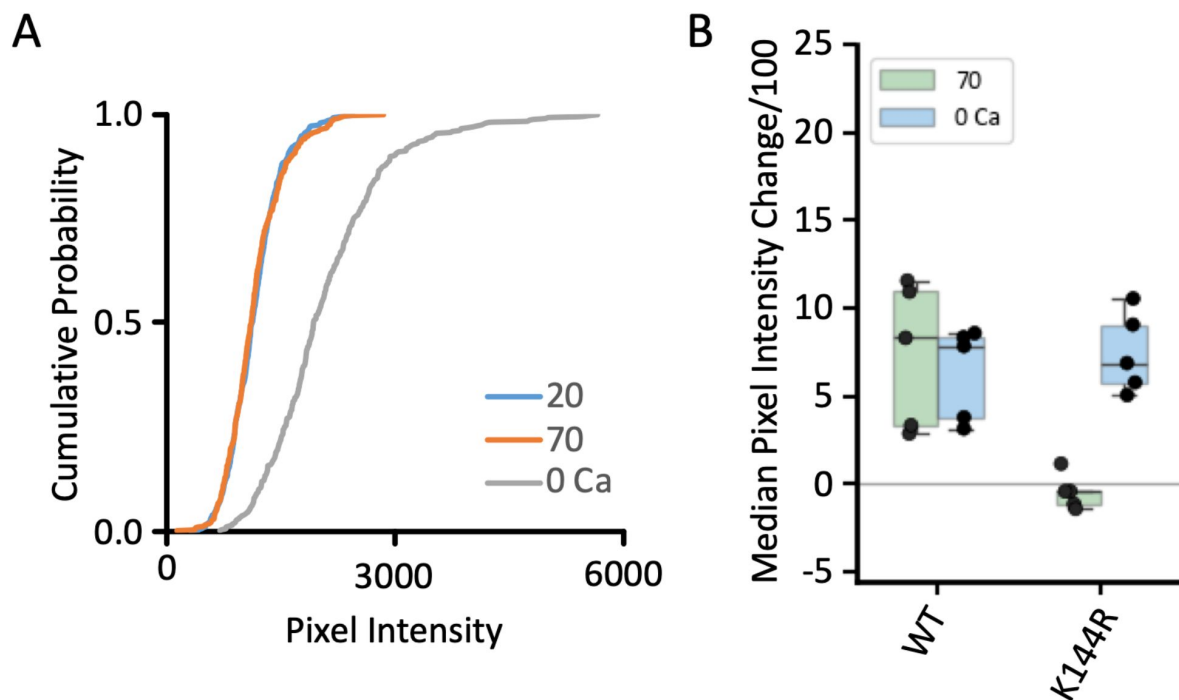


Figure 3, figure supplement 1

Cx43^{K144R} hemichannels are insensitive to changes in PCO₂.

(A) Cumulative probability plot from 5 independent transfections showing dye loading in response to 20 nd 70 mmHg and the zero Ca²⁺ stimulus. (B) Summary graph showing the change in median fluorescence intensity caused by 70 mmHg and zero Ca²⁺ in Cx43^{WT} (data recalculated from Fig 2) and Cx43^{K144R}.

hemichannel function, we used high K^+ induced depolarisation as a way of opening the hemichannel independently of changes in PCO_2 as a positive control. This stimulus resulted in robust ATP release.

CO₂ sensitivity of Cx43 involves multiple Lys residues

As single Lys mutations mostly had partial effects on the CO₂ sensitivity of Cx43 hemichannels, we next mutated pairs of Lys residues. Using the dye loading assay we found that all combinations of paired mutations abolished CO₂ sensitivity relative to the zero Ca^{2+} control (**Fig 5**): K109Q, K144Q (MW test $p = 0.008$, $n=5$); K105Q, K144Q (MW test $p = 0.004$, $n=5$); K105Q, K109Q (MW test $p = 0.004$, $n=5$); K105Q, K234Q (MW test $p = 0.004$, $n=5$); and K144Q, K234Q (MW test $p = 0.004$, $n=5$). Using the GRAB_{ATP} assay we confirmed that all of these double mutations also abolished CO₂-dependent ATP release (MW test for all mutations, CO₂ vs high K^+ positive control $p < 10^{-6}$, **Fig 6**).

The effect of introducing negatively charged residues into the carbamylation motif

We next sought to mimic carbamylation by mutating Lys to Glu. This introduces into the motif the negative charge that would occur upon carbamylation of Lys. In the dye loading assay the mutation K144E resulted in a complete loss of CO₂ sensitivity while dye loading to zero Ca^{2+} was retained (MW test, 70 mmHg vs zero Ca^{2+} : $p = 0.008$, $n=5$, **Fig 7B**). The mutation K105E seemed to disrupt both the CO₂-dependent and the zero Ca^{2+} evoked dye loading (**Fig 7A**). K234E displayed impaired CO₂-dependent dye loading compared to that evoked by zero Ca^{2+} (MW test, 70 mmHg vs zero Ca^{2+} : $p = 0.048$, $n=5$, **Fig 7C**).

Using the GRAB_{ATP} assay, we found that each of these single Lys to Glu mutations disrupted CO₂-dependent ATP release, but did not prevent depolarisation evoked ATP release (**Fig 8**). This disruptive effect on CO₂ dependence might arise because introduction of the negative charge changes the local environment and makes it harder to carbamylate the other Lys residues.

As a change in PCO_2 would likely carbamylate more than one Lys residue simultaneously, we made the double mutation K105E and K109E. With the dye loading assay we found that even at a PCO_2 of 20 mmHg, the cells loaded with dye suggesting that the mutated hemichannels were constitutively open (**Fig 9A, D**). Despite this cells that expressed this double mutant had normal morphology (**Fig 9, movie supplement 1**) and their cultures did not exhibit noticeably higher levels of cell death. Presumably the HeLa cells were able to adapt to the presence of these open channels.

Increasing PCO_2 to 70 mmHg gave no further dye loading. To test whether the channels were open even at a PCO_2 of 20 mmHg, we used the general hemichannel blocker La^{3+} (100 μ M) to show that this reduced the dye loading at 20 mmHg (**Fig 9B, D**, MW test: $p = 0.016$, $n=5$). We used the GRAB_{ATP} assay to demonstrate that at a PCO_2 of 20 mmHg, the K105E, K109E mutation caused continual release of ATP that could be blocked by La^{3+} (**Fig 9C, D**, MW test 20 mmHg vs 20 mmHg + La^{3+} , $p=7.7 \times 10^{-7}$, $n=16$; **Fig 9, movie supplement 1**). This continual release of ATP at a PCO_2 of 20 mmHg was not evident from cells expressing WT Cx43, as application of 100 μ M La^{3+} had no effect on the GRAB_{ATP} fluorescence (**Fig 9C, D**).

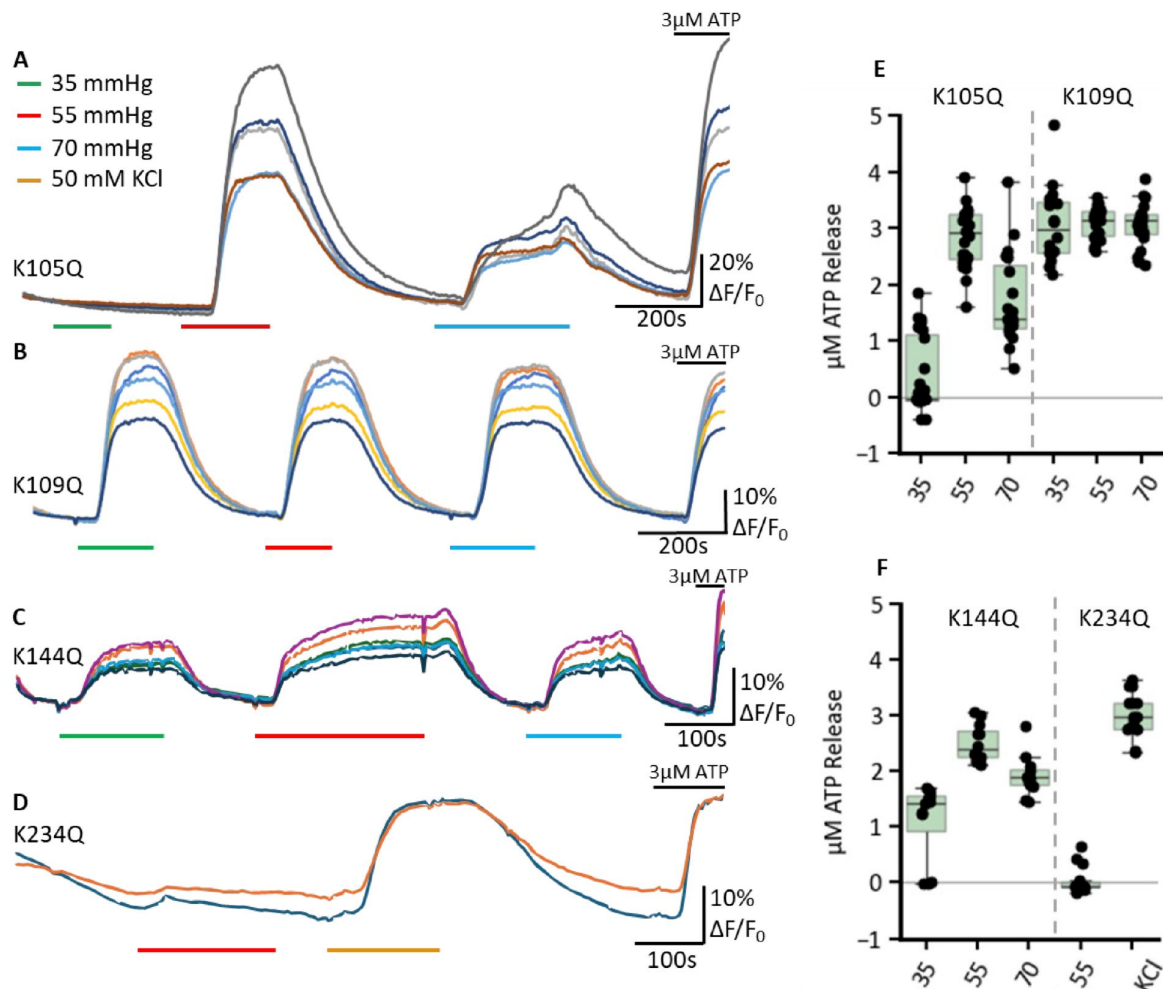


Figure 4

The effect of single Lys mutations on CO₂-dependent ATP release from Cx43.

The genetically-encoded ATP sensor GRAB_{ATP} was co-transfected alongside Cx43 mutant variations, K105Q (A) n=23, K109Q (B) n=20, K144Q (C) n=12, K234Q (D) n=15. Representative trace measurements were selected; each line represents a different cell; the fluorescence was normalised by dividing all values by the baseline median pixel intensity before plotting. Box plot on the right (E) and (F) represent the μM ATP released as determined through normalisation to the 3 μM ATP application.

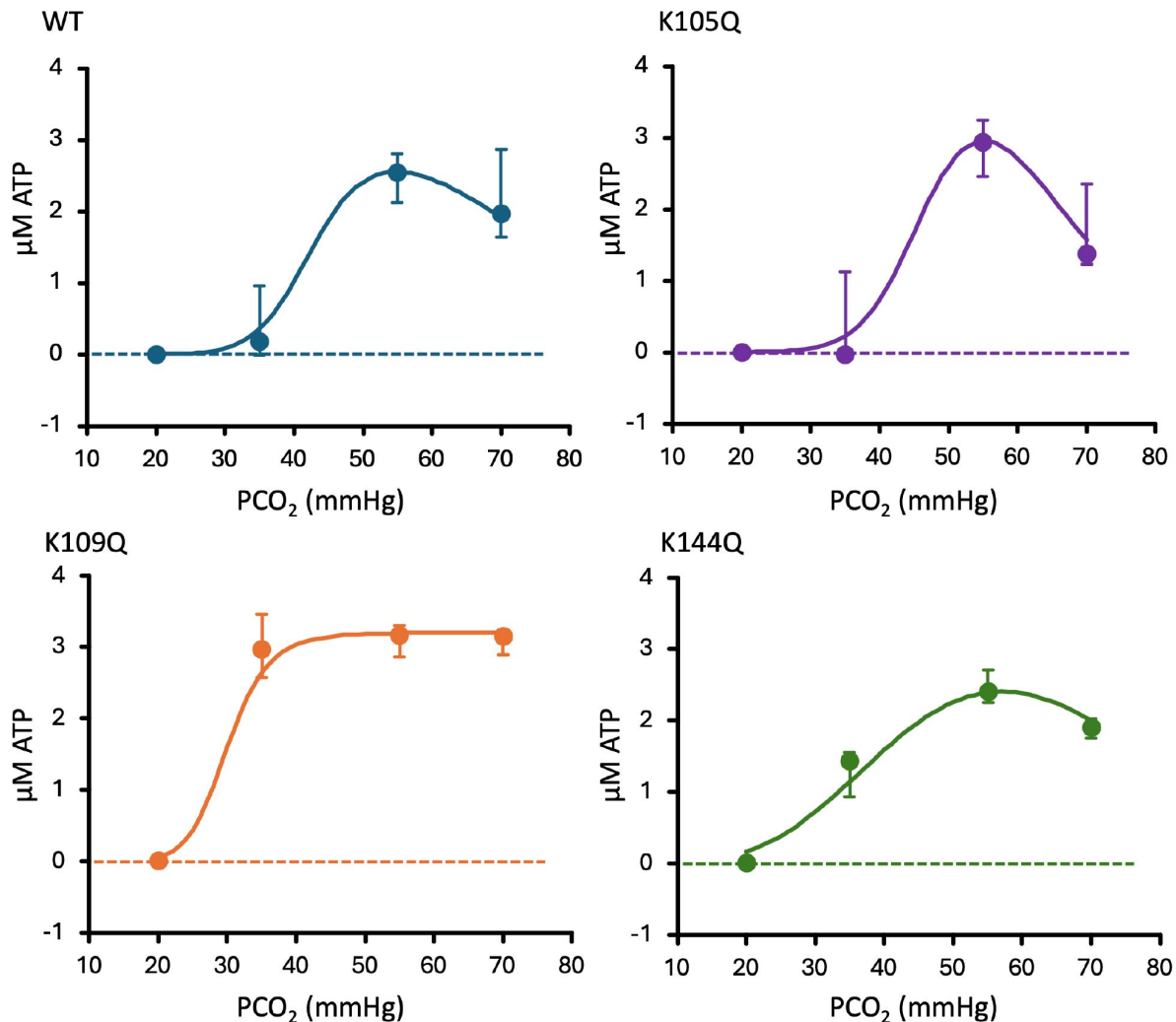


Figure 4, figure supplement 1

The effect of single Lys mutations on the CO₂ dose-response properties of Cx43.

Comparison of Cx43^{WT} (data replotted from [Fig 2C](#)), Cx43^{K105Q}, Cx43^{K109Q} (data replotted from [Fig 4E](#)) and Cx43^{K144Q} (data replotted from [Fig 4F](#)). Data is plotted as medians with lower and upper quartiles. For the WT, K105Q and K144Q the fitted curve is drawn according to a modified Hill equation: $[ATP] = \text{Max}_{ATP} \cdot \frac{(\text{PCO}_2/K)^H}{(1 + (\text{PCO}_2/K)^H)} \cdot \frac{1 - (\text{PCO}_2/K_i)^{H_i}}{1 + (\text{PCO}_2/K_i)^{H_i}}$ Where K and H are the affinity and Hill coefficient of the channel for opening by CO₂, K_i and H_i are the affinity and Hill coefficient for inhibition of the channel by CO₂, and Max_{ATP} is the asymptotically maximum release of ATP to CO₂. For K109Q which did not exhibit CO₂-dependent inhibition at high levels of PCO₂, the Hill equation was used. The parameters for the curves are:

	Cx43 ^{WT}	Cx43 ^{K105Q}	Cx43 ^{K109Q}	Cx43 ^{K144Q}
Max _{ATP} (μM)	3.2	4.5	3.2	3.8
K (mmHg)	43	47	30	43
H	10	10	10	4
K _i (mmHg)	75	65	-	75
H _i	6	8	-	6

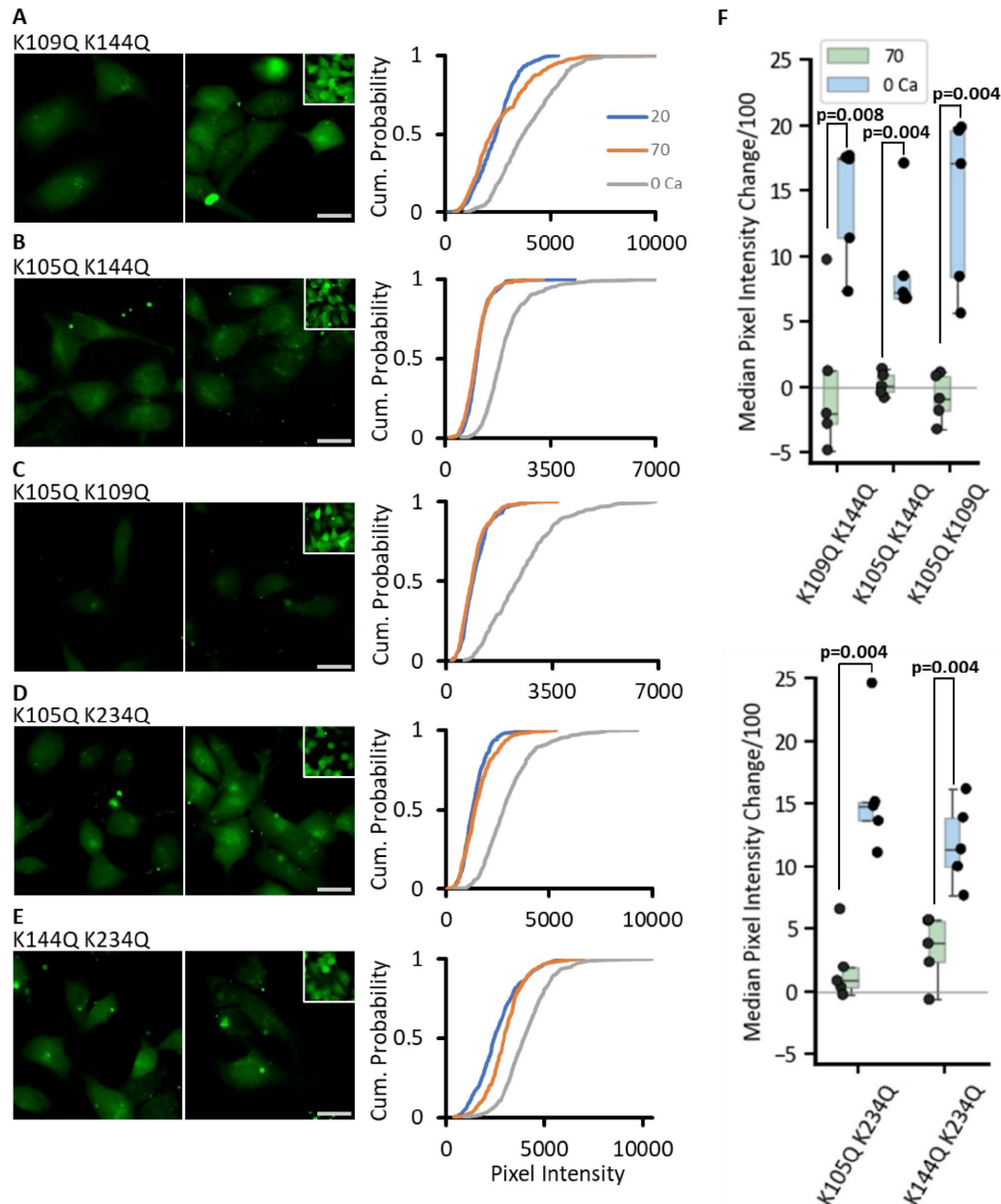


Figure 5

Paired Lys mutations are required to abolish the CO₂ sensitivity of Cx43.

Representative images for all combinations tried are shown – K109Q K144Q (**A**), K105Q K144Q (**B**), K105Q K109Q (**C**), K105Q K234Q (**D**), K144Q K234Q (**E**). Insets represent the 0 Ca²⁺ positive control. For each construct, the pixel intensity in expressing cells was measured from 5 individual transfections, with at least 40 cells per condition. Cumulative probability plots display all measured data points. (**F**) The box plots shows the change in median pixel intensity from 20 mmHg PCO₂ for each transfection for 70 mmHg PCO₂ (green) and 0 Ca²⁺ (blue).

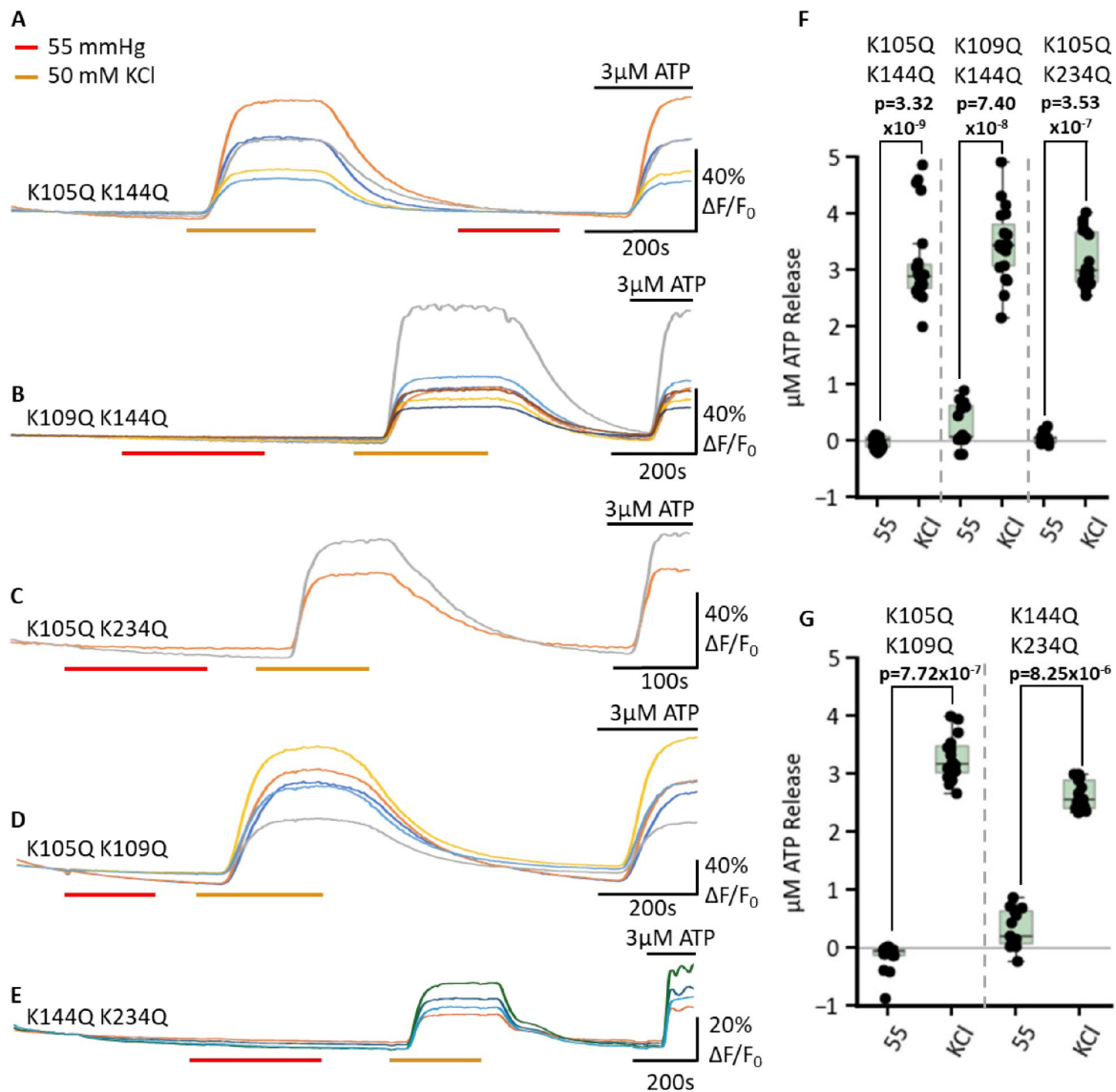


Figure 6

Paired Lys mutations abolish CO₂ dependent ATP release via Cx43.

HeLa cells co-expressing Cx43 and GRAB_{ATP} were subjected to changing PCO₂-levels and fluorescence was recorded. Representative traces (**A-E**) for each of the Cx43 mutations – K105Q K144Q (**A**) n=23, K109Q K144Q (**B**) n=19, K105Q K234Q (**C**) n=17, K109Q K234Q (**D**) n=16, K144Q K234Q (**E**) n=13. The traces indicate normalised fluorescence changes ($\Delta F/F_0$). 3 μM ATP was applied at the end of each experiment to confirm sensor functionality. Furthermore, as a positive control, 50 mM KCl was applied to depolarise the cells and confirm channel function. (**F-G**) Box plots summarising the total ATP release in μM for each double mutant in 55 mmHg PCO₂ and 50 mM KCl. Data points represent individual measured cells.

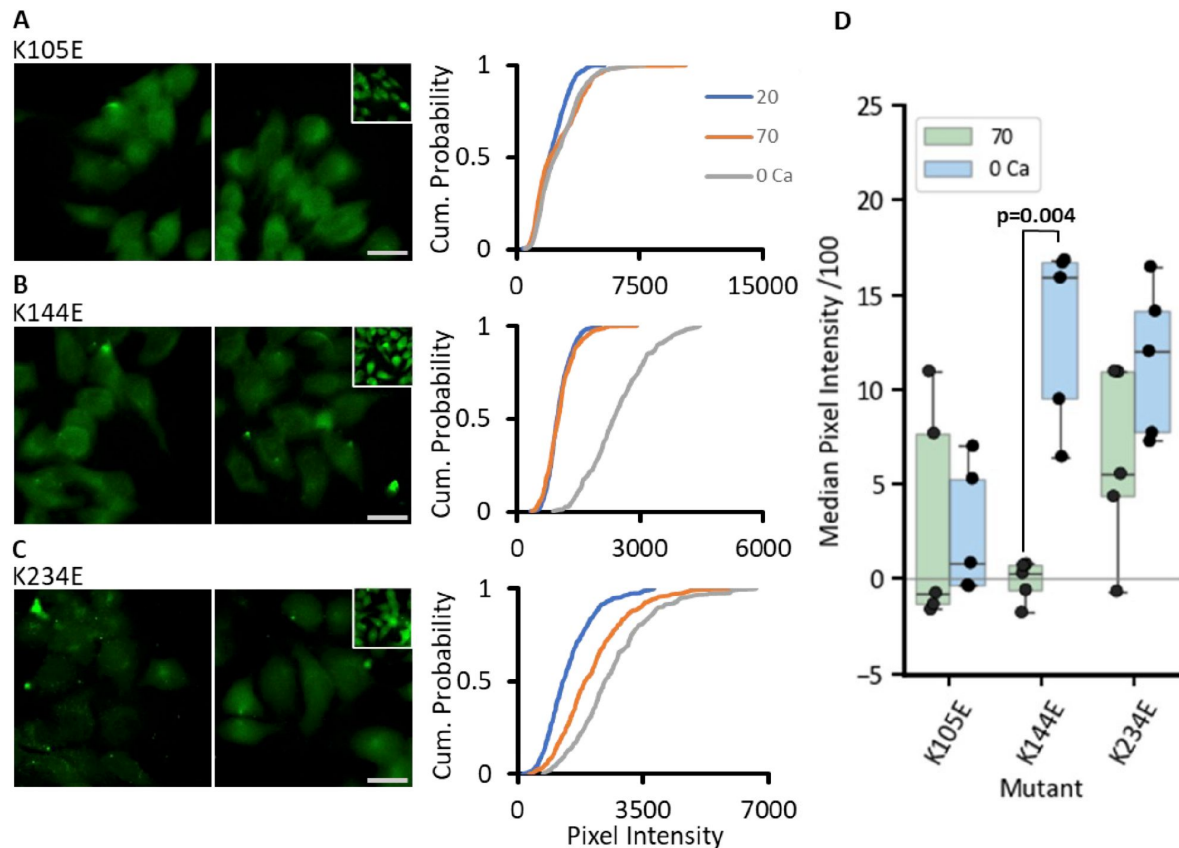


Figure 7

Introduction of negative charge into the carbamylation motif via Lys to Glu mutations has mixed effects on CO₂ sensitivity of Cx43 hemichannels.

Expressing cells pixel intensity for each construct was measured from 5 individual transfections, with at least 40 cells per condition. (A-C) Representative cell images for 20 (left), 70 (right) with insets displaying the 0 Ca²⁺ control for each mutant K105E (A), K144E (B), K234E (C). Scale bar represents 20 μ m. Cumulative probability graphs of pixel intensities are shown on the right for each mutant with three conditions 20 mmHg PCO₂ (blue line), 70 mmHg PCO₂ (orange line) and 0 Ca²⁺ (grey line). The box plot shows change in median pixel intensity from 20 mmHg PCO₂ for each transfection for 70 mmHg PCO₂ (green boxes) and 0 Ca²⁺ (blue boxes).

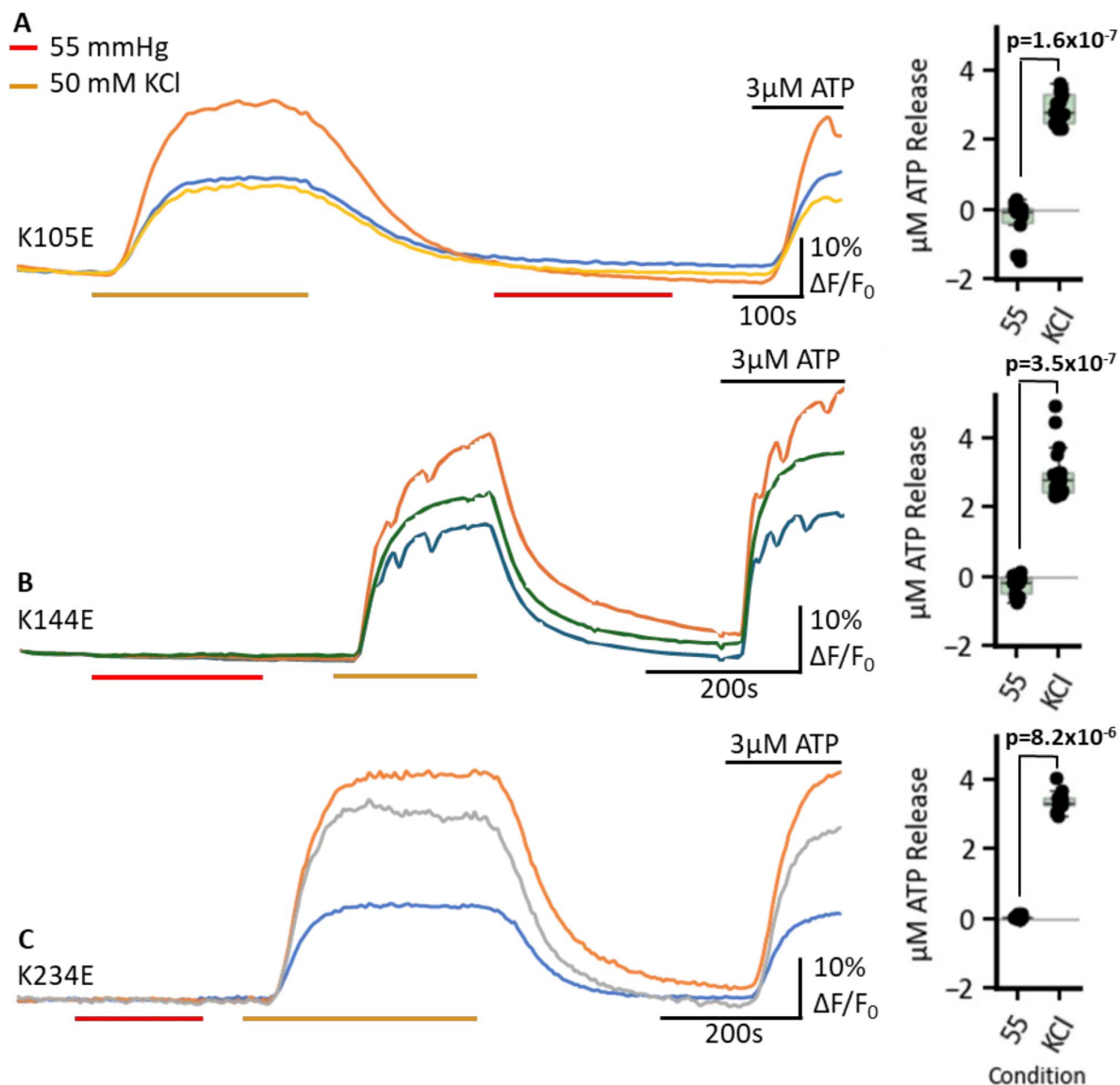


Figure 8

Lys to Glu mutations abolish CO₂-dependent ATP Release.

GRAB_{ATP} fluorescence traces of ATP release from cells expressing single mutant connexins: K105E (**A**) $n=18$, K144E (**B**) $n=17$, K234E (**C**) $n=13$, in response to 55 mmHg PCO₂ (red bars), 50 mM KCl depolarisation control (orange bars) and a 3 μM ATP control application at the end of all experiments to confirm sensor functionality. Traces show changes in normalised fluorescence over time ($\Delta F/F_0$), indicating ATP release. On the right, the box plots display the quantified ATP release in μM for each mutant under 55 mmHg PCO₂ and 50 mM KCl. Each data point represents a measurement from an individual cell.

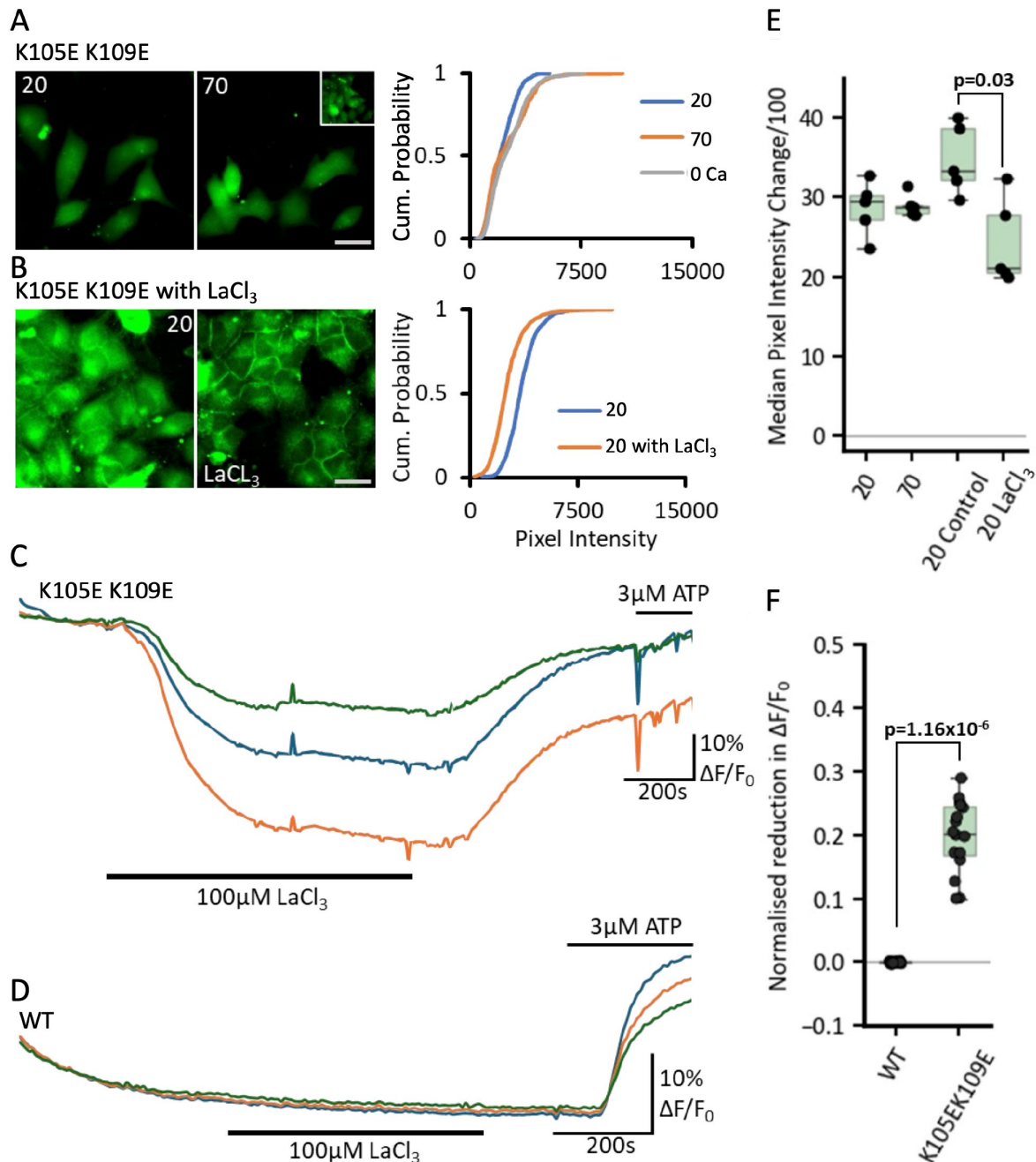


Figure 9

The Cx43 double mutant, K105E K109E, is constitutively open.

(A) Representative fluorescence images of cells expressing the Cx43 K105E K109E mutant under low (20 mmHg PCO₂, left) and high CO₂ (70 mmHg PCO₂, right), inset shows the 0 Ca²⁺ control. Scale bar is 20 μm. Cumulative probability plot of pixel intensity for each condition is shown on the right, overall indicating higher baseline fluorescence and perpetually open hemichannels. (B) Dye-loading with 20 mmHg PCO₂ (left) and 20 mmHg PCO₂ with 100 μM LaCl₃. Cumulative probability plot for pixel intensity under these conditions is shown on the right. (C) Fluorescence traces of ATP release from cells co-expressing GRAB_{ATP} and Cx43 K105E K109E (n=16) under 20 mmHg PCO₂ and 20 mmHg PCO₂ with 100 μM LaCl₃ (black bar). Fluorescence is normalised to baseline. (D) Fluorescence traces of ATP release from cells co-expressing GRAB_{ATP} and Cx43 WT under 20 mmHg PCO₂ and 20 mmHg PCO₂ with 100 μM LaCl₃ (black bar). (E) Box plots summarizing median pixel intensity under the different conditions, showing a significant reduction in intensity in the presence of LaCl₃ (p = 0.03). (F) Box plot shows normalised fluorescence changes values for the difference between - 20 mmHg PCO₂ (representing baseline normalisation) and the application of LaCl₃ for both the K105E K109E and WT Cx43 constructs.

Physiological role for CO₂-dependent

ATP release via Cx43 from astrocytes

Cx43 hemichannels are CO₂ sensitive and are likely to be partially open at typical levels of PCO₂ in mammalian tissue. They could thus be able to release ATP under physiological resting conditions (35 mmHg PCO₂, [Fig 2](#)). Considering the ubiquitous distribution of Cx43 in astrocytes ([de Ceglia et al., 2023](#)) and the likely resting PCO₂ in the brain ([Hogg et al., 1984](#)), we postulated that, barring external modulatory influences, Cx43 hemichannels could continually release low levels of ATP in the hippocampus. Indeed, there exists compelling evidence for astrocytic Cx43 hemichannels being partially open under resting physiological conditions *in vitro* in the hippocampus ([Chever et al., 2014](#)). In this study when Cx43 hemichannels were blocked by the mimetic peptide, Gap26, there was a decrease synaptic strength of about 30% that was occluded by targeted knock out of astrocytic Cx43 expression ([Chever et al., 2014](#)).

As many investigators state that Cx43 hemichannels are shut under physiological conditions, we tested whether the results of [Chever et al \(2014\)](#) could be explained by the CO₂ sensitivity of Cx43. We predicted that if PCO₂ were lowered sufficiently to largely close Cx43, when PCO₂ was returned to the physiological norm, Cx43 should re-open and result in an increase of synaptic strength, presumably downstream of Cx43-dependent ATP release ([Chever et al., 2014](#)). Because ATP can be rapidly broken down to adenosine in the extracellular space ([Frenguelli et al., 2007](#); [Wall & Dale, 2013](#)), which could then act at A1 receptors to inhibit transmission and give confounding effects, we used 8-CPT (8-cyclopentyltheophylline) to selectively block A1 receptors.

We preincubated hippocampal slices in 20 mmHg PCO₂ aCSF for 30 mins prior to recording field EPSPs (fEPSPs) in area CA1 evoked by stimulation of the Schaffer collaterals. At constant extracellular pH, in the presence of 8-CPT, elevation of PCO₂ from 20 to 35 mmHg resulted in a 26% increase in the magnitude of the fEPSP (n=6, [Fig 10A](#)). To test whether this effect was mediated via CO₂-dependent opening of Cx43, we used 100 μM Gap26 to see whether this blocked the increase in fEPSP induced by 35 mmHg PCO₂. Once again the fEPSP increased by about 30% on transfer to 35 mmHg aCSF ([Fig 10B](#)), and application of Gap26 reduced the fEPSP to below its baseline value ([Fig 10B, D](#), [Friedmann 2-way ANOVA: p=0.006, n=6](#)). This was an effect specific to the Gap26 peptide as 100 μM of the scrambled peptide did not have significant effects on the amplitude of the fEPSP ([Fig 10C, D](#), [Friedmann 2-way ANOVA: p=0.55, n=5](#)).

Pathological mutations of Cx43 cause loss of CO₂ sensitivity

A prominent condition associated with mutations of Cx43 is occulodental digital dysplasia (ODDD) ([Laird, 2014](#)). This syndrome involves a range of conditions such as soft tissue fusion of the digits, abnormal craniofacial bone development, small eyes and loss of tooth enamel. Later in age conditions such as glaucoma, skin disease and neuropathies may become evident. As pathological mutations of other CO₂ sensitive connexins alter their CO₂ sensitivity ([Meigh et al., 2014](#); [de Wolf et al., 2016](#); [Cook et al., 2019](#); [Butler & Dale, 2023](#)), we studied two pathological mutations of Cx43. The mutation L90V causes ODDD ([Shibayama et al., 2005](#); [Lai et al., 2006](#)) and A44V causes the skin condition, erythrokeratoderma variabilis et progressiva (EKVP) ([Cocozzelli & White, 2019](#); [Srinivas et al., 2019](#)). We found that the L90V mutation removed CO₂ sensitivity from Cx43 by both the dye loading assay and the GRAB_{ATP} assay ([Fig 11](#)). The mutation A44V blocked CO₂ dependent ATP release ([Fig 11](#)).

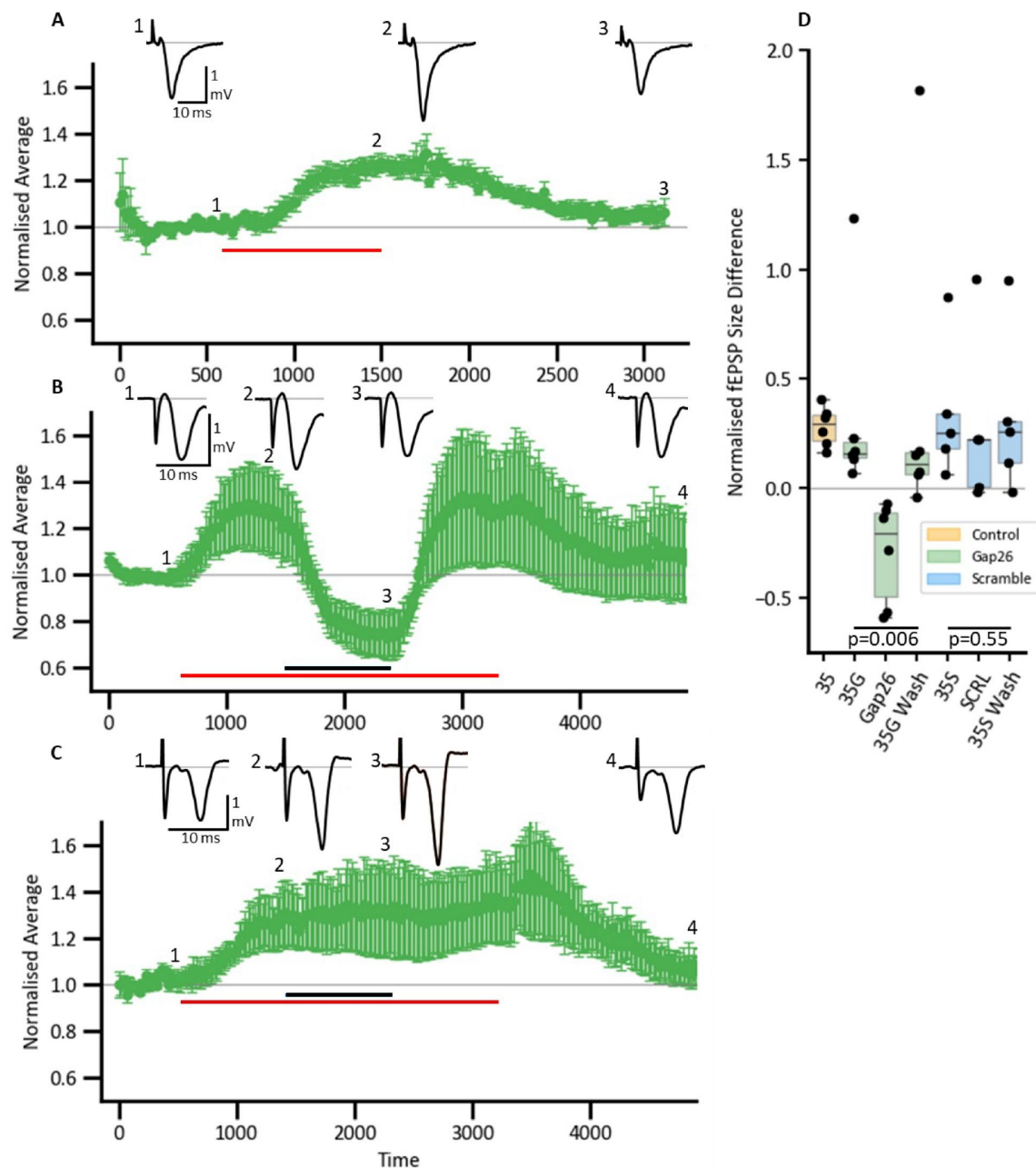


Figure 10

PCO₂-dependent modulation of amplitude of fEPSPs in hippocampus is mediated via Cx43.

(A–C) Time-course plots showing the average amplitude of the normalised fEPSP (± SEM) amplitude in response to different conditions. Inserts display representative fEPSPs. (A) Control condition showing an increase in fEPSP amplitude in response to a modest change to PCO₂ (20 to 35 mmHg), red bar represent the application of 35 mmHg PCO₂, baseline conditions – 20 mmHg PCO₂. (B) EPSP responses in the presence of Gap26 (black bar) and subsequent wash. (C) EPSP responses with scrambled Gap26 peptide applied (black bar). (D) Box plots representing the normalised fEPSP size difference (baseline was subtracted) with colour coded conditions: orange – control 35 mmHg PCO₂, green – Gap26 with pre (35G), after (35S Wash).

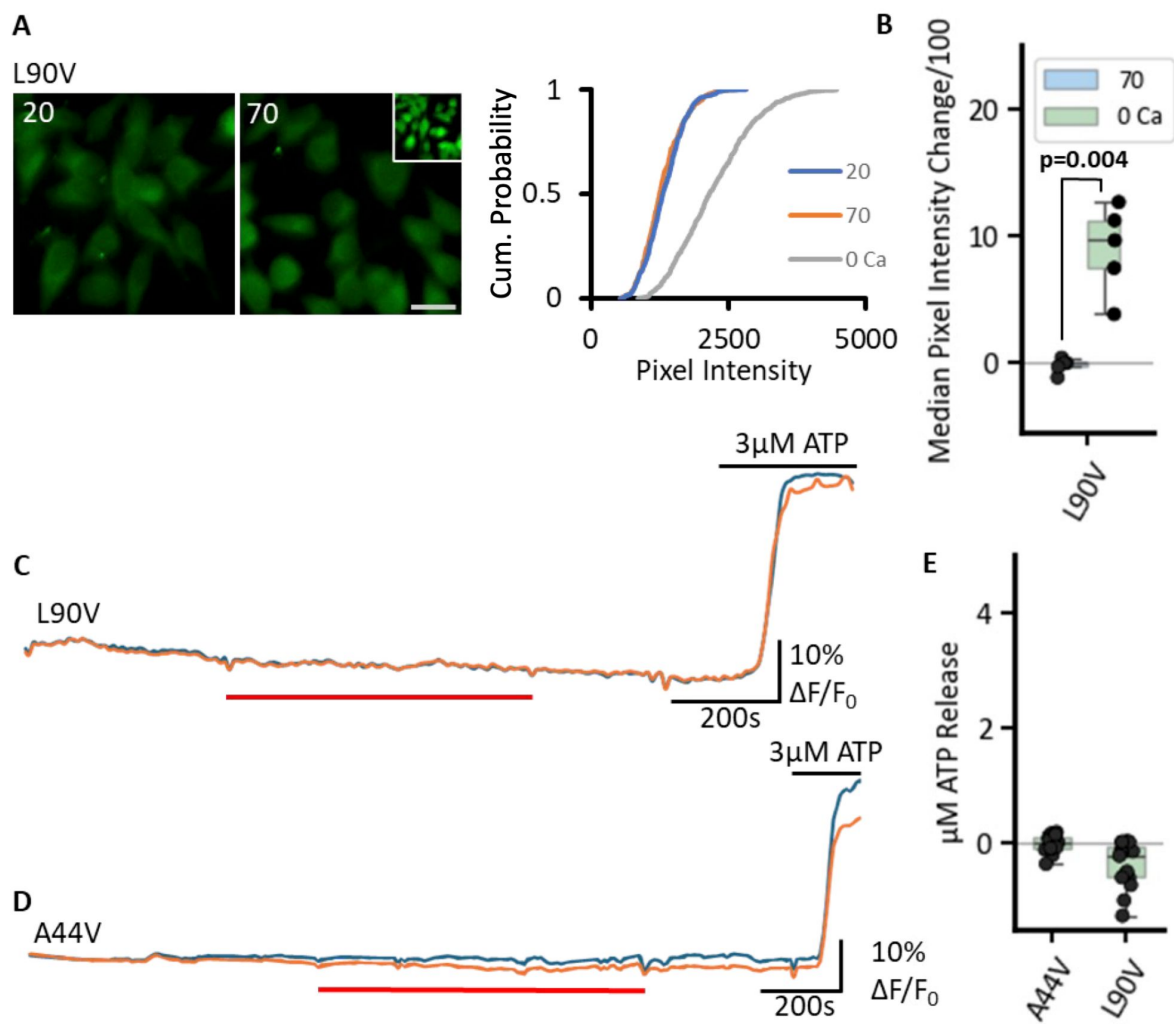


Figure 11

Pathological mutations of Cx43 remove its sensitivity to CO₂.

A,B) L90V prevents CO₂-dependent dye loading. The dye loading assay shows no change in fluorescence between 20 and 70 mmHg, yet functional channels are expressed as shown by the zero Ca²⁺ positive control (inset). Box plot shows the change in median pixel intensity from 20 mmHg PCO₂ to 70 mmHg PCO₂ (green boxes) and 0 Ca²⁺ (blue boxes) for each transfection. **C-E)** GRAB_{ATP} recordings shown that L90V and A44V also abolish CO₂-dependent ATP release (baseline 20 mmHg PCO₂, red bar 70 mmHg PCO₂).

Discussion

Hemichannels of Cx43 are CO₂ sensitive

In contrast to Cx26 expression, which is restricted to specific, small regions of the brain (Nagy *et al.*, 2001 [↗](#); Rash *et al.*, 2001 [↗](#); Nagy *et al.*, 2011 [↗](#)), Cx43 serves as the principal astrocytic connexin, exhibiting widespread presence throughout the brain (Schulz *et al.*, 2015 [↗](#); Yin *et al.*, 2018 [↗](#); de Ceglia *et al.*, 2023 [↗](#)). This widespread distribution, combined with our results, suggests that Cx43 gives the capacity for CO₂ sensitivity across the entire brain.

Our data from three independent assays (dye loading, measurement of ATP release and whole cell recordings) show that hemichannels of Cx43 open to modest changes in PCO₂. However, Cx43 gap junction channels are insensitive to these same changes in PCO₂. At physiological levels of PCO₂ Cx43 hemichannels are partially open and able to release ATP. This observation potentially resolves a puzzling inconsistency in the literature. Many biophysical recordings of Cx43 hemichannels, performed in simplified buffered salines without appreciable dissolved CO₂ or HCO₃⁻, document Cx43 hemichannels as being shut (Contreras *et al.*, 2003 [↗](#); Schalper *et al.*, 2010 [↗](#)). By contrast, under physiological conditions both *in vitro* and *in vivo*, which necessarily involve CO₂-HCO₃⁻ buffering, the evidence suggests that Cx43 hemichannels remain open to some extent (Chever *et al.*, 2014 [↗](#); Guillebaud *et al.*, 2020 [↗](#); Turovsky *et al.*, 2020 [↗](#)). Continual ATP release from astrocytes via Cx43 has been shown to give a tone of ATP that enhances the strength of synaptic transmission in the hippocampus by about 20% (Chever *et al.*, 2014 [↗](#)). Our data is highly complementary to these observations as we find an enhancement of fEPSP amplitude of about 30% when increasing PCO₂ from 20 to 35 mmHg. It is important to note that our experiments were performed under isohydric conditions and differ from a prior study that examined hypocapnia with simultaneous alkalisation. This manipulation enhanced fEPSP strength through a reduction in extracellular adenosine and consequently less inhibition via A1 receptors (Dulla *et al.*, 2005 [↗](#)).

While our data suggests that Cx43 hemichannels are sensitive to PCO₂ over the range 20-70 mmHg, there is an indication that there may be some CO₂-dependent inhibition of hemichannel opening above 55 mmHg. Slightly less ATP release was consistently seen at a PCO₂ of 70 mmHg compared to 55 mmHg. Furthermore, we cannot be sure that Cx43 hemichannels are completely shut at a PCO₂ of 20 mmHg, as it is not possible to reduce the PCO₂ further without changing to HEPES buffered salines. That Gap26 reduced synaptic transmission below the baseline amplitude recorded in slices bathed in 20 mmHg aCSF suggests that even under this condition there may still be a low level of Cx43 hemichannel gating at least in neural tissue which will continually generate CO₂ via oxidative phosphorylation.

Cx43 gap junction channels and hemichannels are closed by intracellular acidification. This involves interaction between the C-terminus and the cytoplasmic loop (Duffy *et al.*, 2002 [↗](#); Hirst-Jensen *et al.*, 2007 [↗](#)). Similar interactions between the C-terminus and cytoplasmic loop are involved in regulating hemichannel activity stimulated by increases in intracellular Ca²⁺ (Iyyathurai *et al.*, 2018 [↗](#)). However, the CO₂ dependent opening of Cx43 is independent of these mechanisms as it persists when the C-terminus has been deleted.

The residues involved in CO₂ sensitivity and possible mechanisms

By exploiting the new structural data for connexins (Myers *et al.*, 2018 [↗](#); Flores *et al.*, 2020 [↗](#); Lee *et al.*, 2020 [↗](#); Yue *et al.*, 2021 [↗](#); Brotherton *et al.*, 2022 [↗](#); Qi *et al.*, 2022 [↗](#); Lee *et al.*, 2023a [↗](#); Lee *et al.*, 2023b [↗](#); Qi *et al.*, 2023 [↗](#)), the predictive power of AF3 and our understanding of carbamylation and CO₂-dependent gating of Cx26 hemichannels and GJCs (Nijjar *et al.*, 2021 [↗](#);

Brotherton *et al.*, 2024 [↗](#)), we identified a series of residues that could plausibly be involved in mediating the CO₂ sensitivity of Cx43: K144, the equivalent of K125 in Cx26; K105, the analogue of R104 in Cx26; K109, the equivalent of K108 in Cx26; and K234, an equivalent of R216 in Cx26. Both K125 and K108 in Cx26 are known to be carbamylated (Nijjar *et al.*, 2025 [↗](#)). Carbamylation of K125 is essential for hemichannel opening to CO₂, whereas carbamylation of both K125 and K108 is required for GJC closure to CO₂ (Nijjar *et al.*, 2025 [↗](#)). R216 in Cx26 has been tentatively identified as a possible interacting partner for K108 but this could be indirect (Nijjar *et al.*, 2025 [↗](#)). Crucially the spatial arrangements of K144, K105, K109 and K234 in AF3 predictions indicate that these residues are sufficiently close to interact across the subunit boundary.

In Cx26, K125 is proposed to interact with R104 following carbamylation, and interestingly mutation of either K125 or R104 is sufficient to abolish CO₂ dependent hemichannel opening and GJC closure. Mutation of K108 is also sufficient by itself to abolish GJC closure to CO₂ (Nijjar *et al.*, 2025 [↗](#)). By contrast single mutations to Gln of any of the 4 Lys residues identified in Cx43 as the equivalents to those involved in CO₂-dependent gating of Cx26 were unable to completely abolish CO₂ sensitivity. Similarly the mutations K125E and R104E in Cx26 create constitutively open hemichannels. However single Lys to Glu mutations in Cx43 did not have this effect. Our data show that in Cx43 at least 2 of the 4 identified residues must be mutated to completely abolish CO₂ sensitivity or to give constitutively open hemichannels. For example mutation of any two Lys residues to Gln, completely abolishes CO₂ dependent ATP release and dye loading. Conversely the double mutation, K105E and K109E, is required for constitutively open hemichannels.

This suggests that in Cx43 there is some redundancy in the effect of CO₂. One possible interpretation is that two carbamate bridges are formed: one being between K144 and K105 (the direct equivalent of K125-R104 in Cx26); and the second being between K109 and K234. There is certainly some data that support this: the double mutations K109Q-K144Q, K105Q-K234Q and K144Q-K234Q being completely insensitive to CO₂. These double mutations would remove both of the proposed bridges. However our observation that the mutations K105Q-K144Q, K109Q-K234Q also completely remove CO₂ sensitivity is inconsistent with a simple two bridge hypothesis as these double mutations would only affect one of the bridges and should therefore leave the other intact to give some degree of CO₂ sensitivity. Instead, we favour a hypothesis in which any of these four residues could be carbamylated but that they all have potential to interact. Depending on which residues become carbamylated a bridge could form between K105 and K234 or between K109 and K144, and possibly even K144 and K234. There is some support for this idea from AF3 as K105 and K109 are only one turn distant in an alpha helix and both could therefore interact with either K144 or K234 following carbamylation. The residues in Cx43 that are involved in CO₂ sensitivity are equivalent to those identified in Cx26, where they participate in both hemichannel and gap junction channel gating to CO₂. However in Cx43 these residues are only involved in mediating the sensitivity of the hemichannel to CO₂. In this respect it is interesting that mutations of the different Lys residues alter the dose-sensitivity of Cx43 to CO₂ in different ways (**Fig 4, figure supplement 1** [↗](#)). New cryo-EM studies of Cx43 hemichannels at vitrified different levels of PCO₂ would shed more light on the mechanism of opening,

Physiological implications of the CO₂ sensitivity of Cx43

Cx43 is the most widely expressed connexin in the human body, being present in every organ system (Lucaciu *et al.*, 2023 [↗](#)). In metabolically highly active organs such as liver, kidney and brain, our data suggest that the physiological PCO₂ will be sufficient to substantially open Cx43 hemichannels (Hogg *et al.*, 1984 [↗](#)). In the context of the brain, Cx43 is the main astrocytic connexin and is expressed in all subtypes of astrocytes (de Ceglia *et al.*, 2023 [↗](#)). This would imply that CO₂-sensitivity mediated via Cx43 will extend to potentially all brain regions. As astrocytes are non-excitable, they are unlikely to become sufficiently depolarised for Cx43 hemichannels to open via voltage dependent gating. We suggest therefore, in non-excitable cells such as astrocytes, that variations in PCO₂ may be the most important physiological regulator of Cx43 hemichannel gating.

For astrocytes, CO₂-dependent opening of Cx43 hemichannels is likely to result in the release of a mix of small, neurochemically significant molecules such as ATP, glutamate, D-serine and lactate, depending on their intracellular concentration.

Cx43 is also highly expressed in the heart, if a proportion of the population were to be in hemichannel form as opposed to GJs (De Smet *et al.*, 2021 [↗](#)), the newly-discovered CO₂ sensitivity of hemichannels may have profound implications for heart function given that cardiomyocytes are highly active, and will thus generate high levels of CO₂ through oxidative phosphorylation. There is a long-established link between CO₂ and cardiac function with effects such as increased coronary blood flow in response to CO₂ as well as changes in myocardial contractile function (Crystal, 2015 [↗](#)). CO₂ can even cause arrhythmias (Zhang *et al.*, 2019 [↗](#)). The effects of CO₂ may be mediated through changes in intracellular pH, which is well known to alter the function of cardiac myocytes (Spitzer *et al.*, 2002 [↗](#); Vaughan-Jones *et al.*, 2009 [↗](#); Orłowski *et al.*, 2025 [↗](#)). However, given that Cx43 is directly sensitive to CO₂, an investigation as to whether CO₂ also has direct effects on cardiac function that are independent of consequent changes in intracellular pH seems warranted.

NF-kappa B and innate immunity exhibit sensitivity to CO₂ (Cummins *et al.*, 2010 [↗](#); Keogh *et al.*, 2017 [↗](#)). Macrophages are CO₂-sensitive, although at least some of this is indirect and via pH changes mediated by carbonic anhydrase activity (Strowitzki *et al.*, 2022 [↗](#)). However, a direct effect of CO₂ on components of the immune system cannot be excluded. In cell culture, when carbonic anhydrase activity is blocked in THP-1 monocytes, the effects of CO₂ are significantly diminished but not abolished, suggesting the existence of a pH-independent pathway (Strowitzki *et al.*, 2022 [↗](#)). As Cx43 is expressed in monocytes it is a plausible candidate to mediate direct CO₂ sensitivity in these cells. Furthermore, Cx43 function in macrophages is deemed critical for various physiological and pathophysiological processes (Rodjakovic *et al.*, 2021 [↗](#)). The role of CO₂ on such pathways remains to be explored.

It is interesting that pathological mutations of Cx43 abrogate CO₂ sensitivity. This follows a pattern seen for Cx26, where a range of mutations linked to syndromic and non-syndromic hearing loss (Meigh *et al.*, 2014 [↗](#); de Wolf *et al.*, 2016 [↗](#); Cook *et al.*, 2019 [↗](#)), and for Cx32, where several mutations that cause X-linked Charcot Marie Tooth Disease (Butler & Dale, 2023 [↗](#)) alter or completely block CO₂ sensitivity. Further studies are needed to see whether loss of CO₂ sensitivity from Cx43 does indeed contribute to pathology.

Evolution of CO₂ sensitivity in connexins

Our previous findings showed that a common carbamylation motif is prevalent across the beta-connexin clade and confers CO₂-sensitivity on the connexins that possess it (Dospinescu *et al.*, 2019 [↗](#)). This motif for example is present in shark Cx32 (which is CO₂ sensitive). Humans and sharks last shared a common ancestor about 400-450 MYA. Our discovery that a very similar carbamylation motif is present in Cx43 and confers CO₂ sensitivity onto their hemichannels. Examining the sequences of other human and non-human alpha connexins reveals that the carbamylation motif with appropriately oriented residues is present in human Cx50, Cx59 and Cx62 as well as Cx40, Cx46 and Cx57 of non-human species (Fig 12 [↗](#), Fig 12 [↗](#) Table Supplement). We have shown Cx50 hemichannels are CO₂ sensitive and this also depends on the carbamylation motif (Lovatt *et al.*, 2025 [↗](#)). These findings suggest that the carbamylation motif and CO₂ sensitivity must have been present in the ancestral connexin gene that predated the establishment of the alpha and beta connexin clades. Further work is needed to understand the evolution of the carbamylation motif and CO₂ sensitivity, but our observations indicate that it is an ancient feature within the molecular phylogeny of the connexins, rather than being a more recently derived feature restricted to the beta connexins.

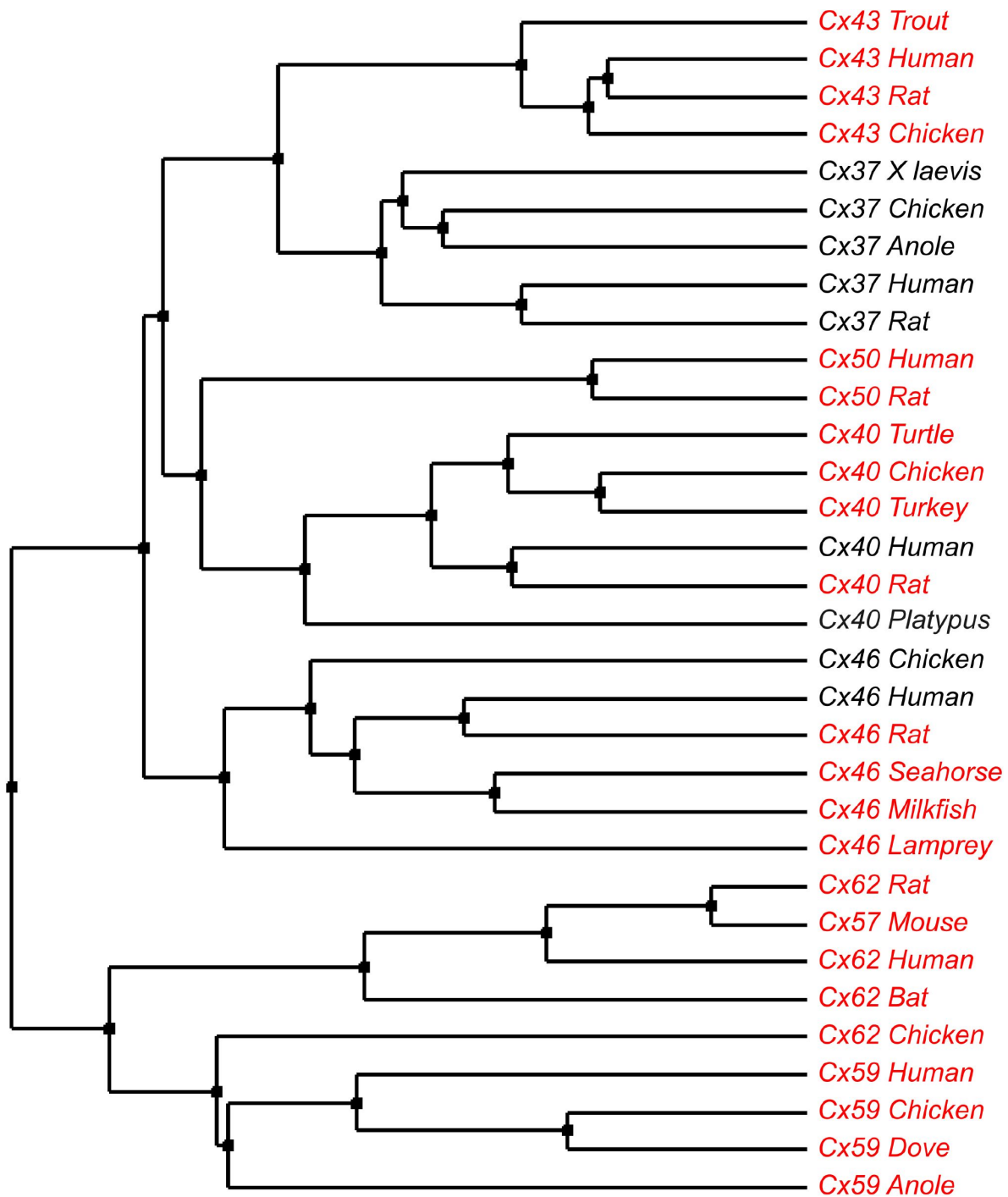


Figure 12

Occurrence of the carbamylation motif in the alpha connexin clade.

Names in red indicate the presence of the motif. The sequences were aligned in TCOFFEE to check for the presence of the motif. The molecular phylogenetic tree was constructed from the aligned traces by the average distance using BLOSUM62 in JalView (Waterhouse *et al.*, 2009 [DOI](#)). Accession numbers for the protein sequences in the tree are given in the accompanying table supplement.

Methods

Recording Solutions

Hypocapnic (20 mmHg PCO₂): 140 mM NaCl, 10 mM NaHCO₃, 1.25 mM NaH₂PO₄, 3mM KCl, 1 mM MgSO₄, 10 mM D-glucose and 2 mM CaCl₂. This was bubbled with a mix of 95%O₂/5%CO₂ and balanced with sufficient pure O₂ from an oxygen concentrator to give a final pH of ~7.3.

Control (35 mmHg PCO₂): 124 mM NaCl, 26 mM NaHCO₃, 1.25 mM NaH₂PO₄, 3 mM KCl, 10 mM D-glucose, 1 mM MgSO₄, 2 mM CaCl₂. This was bubbled with 95%O₂/5% CO₂ and had a final pH of ~7.3.

Hypercapnic (55 mmHg PCO₂): 100 mM NaCl, 50 mM NaHCO₃, 1.25 mM NaH₂PO₄, 3 mM KCl, 10 mM D-glucose, 1 mM MgSO₄, 2 mM CaCl₂. This was bubbled with sufficient CO₂ (~9%, balance O₂) to give a final pH of ~7.3.

Hypercapnic (70 mmHg PCO₂): 73 mM NaCl, 80 mM NaHCO₃, 1.25 mM NaH₂PO₄, 3 mM KCl, 10 mM D-glucose, 1 mM MgSO₄, 2 mM CaCl₂. This was bubbled with sufficient CO₂ (approximately 12%, balance O₂) to give a final pH of ~7.3.

Zero Ca²⁺: 140 mM NaCl, 10 mM NaHCO₃, 1.25 mM NaH₂PO₄, 3mM KCl, 1 mM MgSO₄. On the day of recording, 10 mM D-glucose, 1 mM EGTA (Ethyleneglycol-bis(β-aminoethyl)-N,N',N'-tetraacetic acid) and 2 mM MgCl₂ was added, and the solution bubbled with 95%O₂/5%CO₂ and balanced with sufficient pure O₂ from an oxygen concentrator to give a final pH of ~7.3.

Cloning and mutagenesis

Plasmids containing mutated versions of Cx43 were generated using the Gibson Assembly method (Gibson *et al.*, 2009 [DOI](#)). Overlapping fragments both containing the desired mutation were PCR amplified with primers. Double mutations were cloned using successive Gibson assemblies. PCR fragments were amplified using Q5® High-Fidelity DNA Polymerase (New England Biolabs) (normal PCR or site-directed mutagenesis). The overlap region generated through PCR for Gibson Assembly was 20 base pairs, and the pCAG vector was used as the backbone. All Cx43 constructs were inserted upstream of an mCherry tag, linked via a 12 AA linker (GVPRARDPPVAT). All PCR primers were purchased from Integrated DNA Technologies (IDT). For double mutations, an additional round of cloning was performed using a single mutant as the template, to introduce the additional mutation. For the truncation (Cx43¹⁻²⁵⁶), we created a PCR product of the truncated version of Cx43 and fused it to the same AA linker and mCherry as previously stated, to allow visualisation of cellular localisation. All constructs were confirmed by DNA sequencing (Eurofins GATC sequencing or Plasmidasaurus).

Cell culture

HeLa DH cells were grown in Dulbecco's Modified Eagle Medium (DMEM), supplemented with 10% fetal bovine serum, 50 µg/mL penicillin/streptomycin. HeLa DH cells were used for patch clamp studies on Cx43 and for the dye loading of all Cx43 variants. For dye loading experiments, cells were seeded onto coverslips at a density of 7.5×10^4 cells per well and transiently transfected with the Cx43 constructs following the PEI Transfection Reagent protocol.

Dye-loading

Cells expressing the construct (48-72h) were given an initial 5-minute wash with baseline solution (artificial cerebrospinal fluid – aCSF, 20 mmHg PCO₂). Following this, the cells were exposed to either control solution, hypercapnic (70 mmHg PCO₂), or a zero Ca²⁺ positive control (20 mmHg PCO₂) solution, all of which contained 200 μM 5(6)-carboxyfluorescein (CBF) for 10 min. Next, cells were put into 20 mmHg PCO₂ solution with 200 μM CBF for 5 minutes to close any open channels and prevent dye loss in the next step. Then the coverslip was washed with control solution (without CBF) for another 30 min to remove extracellular dye. For each condition a different coverslip with HeLa cells was used. Cx43 was tagged with mCherry on the C-terminus using a short linker. Expression was verified using mCherry fluorescence. The experiments were replicated using independent transfections 5 times.

Following dye loading, HeLa cells were imaged by epifluorescence (Scientifica Slice Scope, Cairn Research OptoLED illumination, 60x water Olympus immersion objective, NA 1.0, Hamamatsu ImagEM EM-CCD camera, Metafluor software). CBF was excited by a 470 nm LED, with emission captured between 504-543 nm. Cx43 constructs had a C-terminal mCherry tag, which was excited by a 535 nm LED and emission captured between 570-640 nm.

Following acquisition analysis was performed by a person blind to the experimental condition. ImageJ (Wayne Rasband, National Institutes of Health, USA) was used to measure the extent of dye loading by drawing a region of interest (ROI) around each cell, and subsequently, the mean pixel intensity of the ROI was determined. The mean pixel intensity of a representative background ROI for each image was subtracted from each cell measurement from the same image. At least 40 cells were measured for each condition per experiment, and five repetitions of independently transfected HeLa cells were completed. The mean pixel intensities were plotted as cumulative probability distributions, and these graphs show every data point measured. To assess the effect of the 70 mmHg and zero Ca²⁺ solutions on dye loading, the difference in the median pixel intensities between the 70mmHg and 20 mmHg conditions, and the zero Ca²⁺ and 20 mmHg conditions was calculated for each transfection (Figs 3 [↗](#), 5 [↗](#), 7 [↗](#) and 11 [↗](#)).

GRAB_{ATP} recordings

Cells were transiently transfected with the pCAG-Cx43-mCherry construct and pDisplay-GRAB_{ATP}1.0-IRES-mCherry-CAAX (Addgene plasmid # 167582; *RRID:Addgene_167582* [↗](#)) (Wu *et al.*, 2022 [↗](#)) 48 hours prior to imaging. Cells were perfused with control aCSF until a stable baseline was reached, before perfusion with either hypercapnic or high K⁺ aCSF (positive control). Once a stable baseline was reached after solution change, cells were again perfused with control aCSF and when a stable baseline reached, recordings were calibrated by direct application of 3 μM of the corresponding analyte.

All cells were imaged by epifluorescence as above. The cpGFP in GRAB_{ATP} was excited by a 470 nm LED, with emission captured between 504-543 nm. As the Cx43 constructs were mCherry tagged, we selected only cells that expressed both cpGFP and mCherry for recording. GRAB_{ATP} fluorescence images were acquired every 4 seconds. For each condition, at least 3 independent transfections were performed with at least 2 coverslips per transfection.

Analysis of all experiments was carried out in ImageJ. Images were opened as a stack and stabilised. ROIs were drawn around cells co-expressing both sensor and connexin. Median pixel intensity was plotted as normalised fluorescence change ($\Delta F/F_0$) over time to give traces of fluorescence change. Amount of ATP release was quantified as concentration by normalising to the $\Delta F/F_0$ caused by application of 3 μM ATP.

Intracellular pH measurement

BCECF-AM dissolved in DMSO and Pluronic f127 and diluted in 35 mmHg aCSF for a final concentration of 2 μ M. Coverslips seeded with parental HeLa DH cells were incubated in BCECF for 20 minutes, before being washed in 35 mmHg aCSF for 10 minutes. Cells were perfused with control aCSF until a stable baseline was reached, before perfusion with either 55 mmHg or 70 mmHg aCSF. Once a stable baseline was reached after solution change, cells were again perfused with control aCSF. pH_i was then calibrated following the method in (James-Kracke, 1992 [↗](#)). All cells were imaged by epifluorescence with the BCECF excited by a 470 nm LED and emission captured between 504-543 nm. Measurements were obtained from 3 independent transfections were performed with at least 2 coverslips per transfection.

Patch clamp recordings

Coverslips containing non-confluent HeLa cells were placed into a perfusion chamber at room temperature and superfused with control aCSF. An Axopatch 200B amplifier was used to make whole-cell recordings from single HeLa cells. The intracellular fluid in the patch pipettes contained: K-gluconate 60 mM, Cs-gluconate 50 mM, CsCl 10 mM, TEACl 10 mM, EGTA 10 mM, Na₂ATP 3mM, MgCl₂ 3 mM, CaCl₂ 1 mM, HEPES 10 mM, sterile filtered, pH adjusted to 7.2 with KOH. An agarose salt bridge was used to avoid solution changes altering the potential of the Ag/AgCl reference electrode. All whole-cell recordings were performed at a holding potential of -50 mV. Whole-cell conductance was measured by repeated steps to -40 mV. To allow for any drift in whole-cell conductance unrelated to the CO₂ stimulus, the maximal conductance during the CO₂ test stimulus was compared to the average of the conductance before the stimulus and after the stimulus had been fully washed off. Each whole cell recording was considered to be an independent statistical replicate.

Control (35 mmHg) aCSF was used as the baseline and switched to 20 mmHg, 55 mmHg or 70 mmHg aCSF to measure the conductance responses to differing levels of PCO₂. To convert these responses to a PCO₂ dose response curve, assigned a value of zero to 20 mmHg and plotted all changes relative to this. Thus, the values at 35 mmHg were the absolute values of the changes evoked by going from 35 to 20 mmHg, and the absolute value of median change from 35 to 20 mmHg was added to the changes observed going from 35 to 55 and 35 to 70 mmHg.

Imaging assay of gap junction transfer

2-Deoxy-2-[(7-nitro-2,1,3-benzoxadiazol-4-yl)amino]-D-glucose, NBDG, was included at 200 μ M in the patch recording fluid, which contained: K-gluconate 130 mM; KCl 10 mM; EGTA 5 mM; CaCl₂ 2 mM, HEPES 10 mM, pH was adjusted to 7.3 with KOH to give a resulting final osmolarity of 295 mOsm. Cells were imaged on a Cleverscope (MCI Neuroscience) with a Photometrics Prime camera under the control of Micromanager 1.4 software. LED illumination (Cairn Research) and an image splitter (Optosplit, Cairn Research) allowed simultaneous imaging of the mCherry-tagged Cx43 subunits and the diffusion of the NBDG into and between cells. Coupled cells for intercellular dye transfer experiments were initially selected based on tagged Cx43 protein expression and the presence of a gap junctional plaque (**Figure 4, figure supplement 1** [↗](#)). Dye permeation between cells was measured at PCO₂ levels of 20, 55 and 70 mmHg. After establishing the whole cell mode of recording, images were collected every 10 seconds. The assay was performed as described by Nijjar et al (Nijjar *et al.*, 2021 [↗](#)). The start of recording was taken to be the first image following the establishment of a stable whole cell recording. For each gap junction recording (considered as an independent statistical replicate), analysis of the cell images was performed in ImageJ using an ROI drawn on each cell to measure the median pixel intensity of the donor and acceptor cells. The time for the median fluorescence of the acceptor cell to reach 10% of the donor cell was then determined and used as the value for statistical comparisons.

Hippocampal slice preparation

Mice of either sex, 3-5 weeks old, were killed by cervical dislocation and decapitated in accordance with United Kingdom Animals (Scientific Procedures) Act (1986). The brain was dissected and kept on ice both the cerebellum and the rostral section of the brain were removed. 400 μm thick sagittal slices were cut with a vibratome (Microm HM 650V microsliver) in ice-cold cutting solution composed of (in mM: 85 NaCl, 2.5 KCl, 0.5 CaCl_2 , 1.25 NaH_2PO_4 , 24 NaHCO_3 , 25 glucose, and 75 sucrose; pH was adjusted to 7.4, bubbled with 95% O_2 + 5% CO_2). Then slices were incubated in aCSF composed of (in mM: 127 NaCl, 1.9 KCl, 2 MgCl_2 , 2 CaCl_2 , 1.2 KH_2PO_4 , 26 NaHCO_3 , 10 D-glucose, pH 7.4, when bubbled with 95% O_2 and 5% CO_2) at 33 °C. For fEPSP recordings an individual slice was pre-incubated in 20 mmHg PCO_2 aCSF for 30 mins, following which the slice was transferred to a recording chamber submerged in aCSF at 32°C. For stimulation, pulses were delivered by a stimulator via concentric bipolar metal stimulating electrode placed on the surface of CA3. For extracellular recordings, an aCSF-filled microelectrode was placed in CA1.

Statistics

Data is plotted as either cumulative probabilities (showing every data point) or box and whisker plots where the box is interquartile range, bar is median, and whisker extends to most extreme data point that is no more than 1.5 times the interquartile range. All individual data points are superimposed on the plots.

In the case of dye loading one transfection was considered one replicate. In the case of patch clamping and GRAB_{ATP} experiments, one cell was considered one replicate. For dye-loading, to assess the effect a mutation would have on the protein, the difference in dye loading evoked by 70 mmHg PCO_2 or 0 Ca^{2+} conditions (from the 20 mmHg baseline) for each construct were compared, using the median value for each condition from each transfection. The Mann-Whitney U one-sided test was used for the comparison on the basis that *a priori* we expected mutations to either reduce CO_2 sensitivity or have no effect. To compare the fEPSP size across different conditions, the non-parametric Friedman two-way ANOVA was used. Statistical analysis was performed using the python language and SciPy library.

Reagents used

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Chemical compound, drug	DMEM	Merck Life Sciences UK Ltd	CAT# D6046	
Chemical compound, drug	Fetal bovine serum	Labtech.com	CAT# FCS-SA	
Chemical compound, drug	GeneJuice Transfection Reagent	Merck Life Sciences UK Ltd	CAT# 70967-3	
Chemical compound, drug	PEI Prime™ linear polyethylenimine	Merck Life Sciences UK Ltd	CAT# 919012	

Chemical compound, drug	5(6)-Carboxyfluorescein	Merck Life Sciences UK Ltd	CAT# 8510820005	
Chemical compound, drug	Opti-MEM™ I Reduced Serum Medium	Thermo Fisher Scientific	CAT# 31985070	
pDisplay-GRAB_ATP1.0	Plasmid	Addgene	plasmid#167582 ; RRID:Addgene_167582	
Chemical compound, drug	GeneJuice	Merck Sigma-Aldrich	CAT# 70967	
Chemical compound, drug	5(6)-Carboxyfluorescein	Novabiochem	CAT# 8.51082	
Chemical compound, drug	2-Deoxy-2-[(7-nitro-2,1,3-benzoxadiazol-4-yl)amino]-D-glucose	AAT Bioquest	CAT# 36702	
Chemical compound, drug	BCECF, AM (2',7'-Bis-(2-Carboxyethyl)-5-(and-6)-Carboxyfluorescein, Acetoxymethyl Ester)	Thermo Fisher Scientific	CAT# 11524147	
Sequence-based reagent	Cx43-forward	IDT	PCRprimers	TACCGCGGGCCCG GGATCCACCGGTAT GGGTGACTGGAGC GCC
Sequence-based reagent	Cx43-reverse	IDT	PCRprimers	GCGGTACCCCGATC TCCAGGTCATCAGG CC
Sequence-based reagent	mCherry-forward	IDT	PCRprimers	CCTGGAGATCGGG GTACCGCGGGCCCG GG
Sequence-based reagent	mCherry-reverse	IDT	PCRprimers	CTTGATACTTACCT GCGGCCTCGAGTTA CTTGACAGCTCGT CCATGCCGCCG
Sequence-based reagent	K105Q-forward	IDT	PCRprimers	GGAAGAGCAACTGA ACAAGAAAGAGGAA G
Sequence-based reagent	K105Q-reverse	IDT	PCRprimers	TCTTGTTCAAGTTGC TCTTCCTTTCGCAT CACATAG
Sequence-based reagent	K109Q-forward	IDT	PCRprimers	GAACAAGCAAGAGG AAGAACTCAAGGTT GCCC
Sequence-based reagent	K109Q-reverse	IDT	PCRprimers	GTTCTTCTCTTGC TTGTTCAAGTTTCTCT TCCTTTCG
Sequence-based reagent	K144Q-forward	IDT	PCRprimers	GCATGGTCAGGTGA AAATGCGAGGGGG G
Sequence-based reagent	K144Q-reverse	IDT	PCRprimers	GCATTTTCACCTGA CCATGCTCTTCAAT ACCGTAC

Sequence-based reagent	234Q-forward	IDT	PCRprimers	TTTCTTCCAGGGCG TTAAGGATCGGGTT AAGG
Sequence-based reagent	234Q-reverse	IDT	PCRprimers	CCTTAACGCCCTGG AAGAAAACATAGAA GAGTTCAATGATAT TCAAG
Sequence-based reagent	105E-forward	IDT	PCRprimers	GGAAGAGGAACTGA ACAAGAAAGAGGAA G
Sequence-based reagent	105E-reverse	IDT	PCRprimers	TCTTGTTCAAGTTCCT CTTCCTTTGCGATC ACATAG
Sequence-based reagent	144E-forward	IDT	PCRprimers	GCATGGTGAGGTGA AAATGCGAGGGGG G
Sequence-based reagent	144E-reverse	IDT	PCRprimers	GCATTTTCACCTCA CCATGCTCTTCAAT ACCGTAC
Sequence-based reagent	234E-forward	IDT	PCRprimers	TTTCTTCGAGGGCG TTAAGGATCGGGTT AAGG
Sequence-based reagent	234E-reverse	IDT	PCRprimers	CCTTAACGCCCTCG AAGAAAACATAGAA GAGTTCAATGATAT TCAAG
Sequence-based reagent	K105Q forward	K109Q-IDT	PCRprimers	GCAACTGAACAAGC AAGAGGAAGAACTC AAGGTTGCCC
Sequence-based reagent	K105Q reverse	K109Q-IDT	PCRprimers	GTTCTTCCTCTTGC TTGTTCAAGTTGCTC TTCCTTTC
Sequence-based reagent	K105E forward	K109E-IDT	PCRprimers	GGAAGTGAACAAGG AAGAGGAAGAACTC AAGGTTGCCC
Sequence-based reagent	K105E reverse	K109E-IDT	PCRprimers	GTTCTTCCTCTTCTT TGTTCAAGTTCTCTT CCTTTC
Sequence-based reagent	Cx43 ¹⁻²⁵⁶ forward	IDT	PCRprimers	TTT GGC AAA GAA TTC GGT ACC GCG GGC CCG GGA TCC AC
Sequence-based reagent	Cx43 ¹⁻²⁵⁶ reverse	IDT	PCRprimers	GGA TCC CGG GCC CGC GGT ACC CCT TTG GCA GGG CTC AGC GC
Software, algorithm	PyMol		https://pymol.org/	
Software, algorithm	AlphaFold3		https://alphafoldserver.com/	
Software, algorithm	JalView 2.11.5.0		https://www.jalview.org	

Data availability

All data generated in the paper is available as supplements to the figures.

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Additional files

Figure 9 movie supplement 1. *GRAB_{ATP} fluorescence recorded from HeLa cells expressing Cx43^{K105E} during application of 100 μ M LaCl₃ in 20 mmHg PCO₂ aCSF.* [↗](#)

Figure 2—Source data 1. [↗](#)

Figure 3—Source data 1. [↗](#)

Figure 4—Source data 1. [↗](#)

Figure 5—Source data 1. [↗](#)

Figure 6—Source data 1. [↗](#)

Figure 7—Source data 1. [↗](#)

Figure 8—Source data 1. [↗](#)

Figure 9—Source data 1. [↗](#)

Figure 10—Source data 1. [↗](#)

Figure 11—Source data 1. [↗](#)

Figure 12 Supplementary Table. [↗](#)

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Reviewer #1 (Public review):

Summary:

This study builds on previous work demonstrating that several beta connexins (Cx26, Cx30 and Cx32) have a carbamylation motif which renders them sensitive to CO₂. In response to CO₂, hemichannels composed of these connexins open, enabling diffusion of small molecules (such as ATP) between the cytosol and extracellular environment. Here, the authors have identified that an alpha connexin, Cx43, also contains a carbamylation motif, and they demonstrate that CO₂ opens Cx43 hemichannels. Most of the study involves using transfected cells expressing wild-type and mutant Cx43 to define amino acids required for CO₂ sensitivity. Hippocampal tissue slices in culture were used to show that CO₂-induced synaptic transmission was affected by Cx43 hemichannels, providing a physiological context. The authors point out that the Cx43 gene significantly diverges from the beta connexins that are CO₂ sensitive, suggesting that the conserved carbamylation motif was present before the alpha and beta connexin genes diverged.

Strengths:

The molecular analysis defining the amino acids which contribute to the CO₂ sensitivity of Cx43 is a major strength of the study. The rigor of analysis was strengthened by using three independent assays for hemichannel opening: dye uptake, patch clamp channel measurements and ATP secretion. The resulting analysis identified key lysines in Cx43 that were required for CO₂-mediated hemichannel opening. A double K to E Cx43 mutant produced a construct that produced hemichannels that were constitutively open, which further strengthened the analysis.

Using hippocampal tissue sections to demonstrate that CO₂ can influence field excitatory postsynaptic potentials (fEPSPs) provides a native context for CO₂ regulation of Cx43 hemichannels. Cx43 mutations associated with Oculodentodigital Dysplasia (ODDD) inhibited CO₂-induced hemichannel opening, although the mechanism by which this occurs was not elucidated.

Cytosolic pH was measured and it was further demonstrated that Cx43 hemichannels composed of untagged Cx43 are sensitive to CO₂.

A molecular phylogenetic survey was performed which identified several other non-beta connexins that have a putative carbamylation motif. How this relates to connexin evolution was added to the discussion.

Weaknesses:

Cultured cells are typically grown in incubators containing 5% CO₂ which is ~40 mmHg. Determining compensatory mechanisms that enable the cells to be viable if Cx43 hemichannels are open at this PCO₂ would strengthen the study.

Experiments using Gap26 to inhibit Cx43 hemichannels in fEPSP measurements used a scrambled peptide as a control. Including gap peptides specifically targeting Cx26, Cx30 and Cx32 as additional controls would strengthen the study, since the tissue sections have a complex pattern of connexin expression.

<https://doi.org/10.7554/eLife.105989.2.sa3>

Reviewer #2 (Public review):

Summary:

This paper examines the CO₂ sensitivity of Cx43 hemichannels and gap junctional channels in transiently transfected Hela cells using several different assays including ethidium dye uptake, ATP release, whole cell patch clamp recordings and an imaging assay of gap junctional dye transfer. The results show that raising pCO₂ from 20 to 70 mmHg (at a constant pH of 7.3) cause an increase in opening of Cx43 hemichannels but did not block Cx43 gap junctions. This study also showed that raising pCO₂ from 20 to 35 mm Hg resulted in an increase in synaptic strength in hippocampal rat brain slices, presumably due to downstream ATP release, suggesting that the CO₂ sensitivity of Cx43 may be physiologically relevant. As a further test of the physiological relevance of the CO₂ sensitivity of Cx43, it was shown that two pathological mutations of Cx43 that are associated with ODDD caused loss of Cx43 CO₂-sensitivity. Cx43 has a potential carbamylation motif that is homologous to the motif in Cx26. To understand the structural changes involved in CO₂ sensitivity, a number of mutations were made in Cx43 sites thought to be the equivalent of those known to be involved in the CO₂ sensitivity of Cx26 and the CO₂ sensitivity of these mutants was investigated.

Strengths:

This study shows that the apparent lack of functional Cx43 hemichannels observed in a number of previous in vitro function studies may be due to the use of HEPES to buffer the external pH. When Cx43 hemichannels were studied in external solutions in which CO₂/bicarbonate was used to buffer pH instead of HEPES, Cx43 hemichannels showed significantly higher levels of dye uptake, ATP release, and ionic conductance. These findings may have major physiological implications since Cx43 hemichannels are found in many organs throughout the body including the brain, heart and immune system.

Weaknesses:

Interpretation of the site-directed mutation studies is complicated. Although Cx43 has a potential carbamylation motif that is homologous to the motif in Cx26, the results of site-directed mutation studies were inconsistent with a simple model in which K144 and K105 interact following carbamylation to cause the opening of Cx43 hemichannels.

Secondly, although it is shown that two Cx43 ODDD associated mutations show a loss of CO₂ sensitivity, there is no evidence that the absence of CO₂ sensitivity is involved in the pathology of ODDD.

<https://doi.org/10.7554/eLife.105989.2.sa2>

Reviewer #3 (Public review):

In this paper, authors aimed to investigate carbamylation effects on the function of Cx43-based hemichannels. Such effects have previously been characterized for other connexins, e.g. for Cx26, which display increased hemichannel (HC) opening and closure of gap junction channels upon exposure to increased CO₂ partial pressure (accompanied by increased bicarbonate to keep pH constant). The authors used HeLa cells transiently transfected with Cx43 to investigate CO₂-dependent carbamylation effects on Cx43 HC function. In contrast to Cx43-based gap junction channels that are here reported to be insensitive to PCO₂ alterations, they provide evidence that Cx43 HC opening is highly dependent on the PCO₂ pressure in the bath solution, over a range of 20 up to 70 mmHg encompassing the physiologically normal resting level of around 40 mmHg. They furthermore identified several Cx43 residues involved in Cx43 HC sensitivity to PCO₂: K105, K109, K144 & K234; mutation of 2 or more of these AAs is necessary to abolish CO₂ sensitivity. The subject is interesting and the results indicate that a fraction of HCs is open at a physiological 40 mmHg PCO₂, which differs from the situation under HEPES buffered solutions where HCs are mostly closed under resting conditions. The mechanism of HC opening with CO₂ gassing is linked to carbamylation and authors pinpointed several Lys residues involved in this process. Overall, the work is interesting as it shows that Cx43 HCs have a significant open probability under resting conditions of physiological levels of CO₂ gassing, probably applicable to/relevant for brain, heart and other Cx43 expressing organs. The paper gives a detailed account on various experiments performed (dye uptake, electrophysiology, ATP release to assess HC function) and results concluded from those. They further consider many candidate carbamylation sites by mutating them to negatively charged Glu residues. The paper finalizes with hippocampal slice work showing evidence for connexin-dependent increases of the EPSP amplitude that could be inhibited by HC inhibition with Gap26 (Fig. 10). Another line of evidence comes from the Cx43-linked ODDD genetic disease whereby L90V as well as the A44V mutations of Cx43 prevented the CO₂ induced hemichannel opening response (Fig. 11). Although the paper is interesting, in its present state it suffers from (i) a problematic Fig. 3, precluding interpretation of the data shown, and (ii) the poor use of hemichannel inhibitors that are necessary to strengthen the evidence in the crucial experiment of Fig. 2 and others.

Comments on revisions:

The traces in Fig.2B show that the HC current is inward at 20 mmHg PCO₂, while it switches to an outward current at 55mmHg PCO₂. HCs are non-selective channels, so their current should switch direction around 0 mV but not around -50 mV. As such, the -50 mV switching point indicates involvement of another channel distinct from non-selective Cx43 hemichannels. In the revised version, this problem has not been solved nor addressed. Additionally, I identified another problem in that the experimental traces shown lack a trace at the baseline condition of PCO₂ 35mmHg, while the summary graph depicts a data point. Not showing a trace at baseline PCO₂ 35mmHg renders data interpretation in the summary graph questionable.

<https://doi.org/10.7554/eLife.105989.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

This study builds on previous work demonstrating that several beta connexins (Cx26, Cx30, and Cx32) have a carbamylation motif which renders them sensitive to CO₂. In response to CO₂, hemichannels composed of these connexins open, enabling diffusion of small molecules (such as ATP) between the cytosol and extracellular environment. Here, the authors have identified that an alpha connexin, Cx43, also contains a carbamylation motif, and they demonstrate that CO₂ opens Cx43 hemichannels. Most of the study involves using transfected cells expressing wildtype and mutant Cx43 to define amino acids required for CO₂ sensitivity. Hippocampal tissue slices in culture were used to show that CO₂-induced synaptic transmission was affected by Cx43 hemichannels, providing a physiological context. The authors point out that the Cx43 gene significantly diverges from the beta connexins that are CO₂ sensitive, suggesting that the conserved carbamylation motif was present before the alpha and beta connexin genes diverged.

Strengths:

(1) The molecular analysis defining the amino acids that contribute to the CO₂ sensitivity of Cx43 is a major strength of the study. The rigor of analysis was strengthened by using three independent assays for hemichannel opening: dye uptake, patch clamp channel measurements, and ATP secretion. The resulting analysis identified key lysines in Cx43 that were required for CO₂-mediated hemichannel opening. A double K to E Cx43 mutant produced a construct that produced hemichannels that were constitutively open, which further strengthened the analysis.

(2) Using hippocampal tissue sections to demonstrate that CO₂ can influence field excitatory postsynaptic potentials (fEPSPs) provides a native context for CO₂ regulation of Cx43 hemichannels. Cx43 mutations associated with Oculodentodigital Dysplasia (ODDD) inhibited CO₂-induced hemichannel opening, although the mechanism by which this occurs was not elucidated.

Weaknesses:

(1) Cx43 channels are sensitive to cytosolic pH, which will be affected by CO₂. Cytosolic pH was not measured, and how this affects CO₂-induced Cx43 hemichannel activity was not addressed.

We have now addressed this with intracellular pH measurements and removal of the C-terminal pH sensor from Cx43 -the hemichannel remains CO₂ sensitive.

(2) Cultured cells are typically grown in incubators containing 5% CO₂, which is ~40 mmHg. It is unclear how cells would be viable if Cx43 hemichannels are open at this PCO₂.

The cells look completely healthy with normal morphology and no sign of excessive cell death in the cultures. Presumably they have ways of compensating for the effects of partially open Cx43 hemichannels.

(3) Experiments using Gap26 to inhibit Cx43 hemichannels in fEPSP measurements used a scrambled peptide as a control. Analysis should also include Gap peptides specifically

| targeting Cx26, Cx30, and Cx32 as additional controls.

We don't feel this is necessary given the extensive prior literature in hippocampus showing the effect of ATP release via open Cx43 hemichannels on fEPSP amplitude that used astrocytic specific knockout of Cx43 and Gap26 (doi: 10.1523/jneurosci.0015-14.2014).

| (4) *The mechanism by which ODDD mutations impair CO₂-mediated hemichannel opening was not addressed. Also, the potential roles for inhibiting Cx43 hemichannels in the pathology of ODDD are unclear.*

These pathological mutations that alter CO₂ sensitivity are similar to pathological mutation in Cx26 and Cx32, which also remove CO₂ sensitivity. Our cryo-EM studies on Cx26 give clues as to why these mutations have this effect -they alter conformational mobility of the channel (Brotherton et al 2022 doi: 10.1016/j.str.2022.02.010 and Brotherton et al 2024 doi: 10.7554/eLife.93686). We assume that similar considerations apply to Cx43, but this requires improved cryoEM structures of Cx43 hemichannels at differing levels of PCO₂.

We agree that the link between loss of CO₂ sensitivity of Cx43 and ODDD is not established and have revised the text to make this clear.

| (5) *CO₂ has no effect on Cx43-mediated gap junctional communication as opposed to Cx26 gap junctions, which are inhibited by CO₂. The molecular basis for this difference was not determined.*

Cx26 gap junction channels are so far unique amongst CO₂ sensitive connexins in being closed by CO₂. We have addressed the mechanism by which this occurs in Nijjar et al 2025 DOI: 10.1113/JP285885 -the requirement of carbamylation of K108 in Cx26 (in addition to K125) for GJC closure.

| (6) *Whether there are other non-beta connexins that have a putative carbamylation motif was not addressed. Additional discussion/analysis of how the evolutionary trajectory for Cx43 maintaining a carbamylation motif is unique for non-beta connexins would strengthen the study.*

We have performed a molecular phylogenetic survey to show that the carbamylation motif occurs across the alpha connexin clade and have shown that Cx50 is indeed CO₂ sensitive (doi: 10.1101/2025.01.23.634273). This is now in Fig 12.

Reviewer #2 (Public review):

Summary:

This paper examines the CO₂ sensitivity of Cx43 hemichannels and gap junctional channels in transiently transfected Hela cells using several different assays, including ethidium dye uptake, ATP release, whole cell patch clamp recordings, and an imaging assay of gap junctional dye transfer. The results show that raising pCO₂ from 20 to 70 mmHg (at a constant pH of 7.3) causes an increase in opening of Cx43 hemichannels but does not block Cx43 gap junctions. This study also showed that raising pCO₂ from 20 to 35 mm Hg resulted in an increase in synaptic strength in hippocampal rat brain slices, presumably due to downstream ATP release, suggesting that the CO₂ sensitivity of Cx43 may be physiologically relevant. As a further test of the physiological relevance of the CO₂ sensitivity of Cx43, it was shown that two pathological mutations of Cx43 that are associated with ODDD caused loss of Cx43 CO₂-sensitivity. Cx43 has a potential carbamylation motif that is homologous to the motif in Cx26. To understand the structural changes involved in CO₂ sensitivity, a number of mutations were made in Cx43

sites thought to be the equivalent of those known to be involved in the CO₂ sensitivity of Cx26, and the CO₂ sensitivity of these mutants was investigated.

Strengths:

This study shows that the apparent lack of functional Cx43 hemichannels observed in a number of previous in vitro function studies may be due to the use of HEPES to buffer the external pH. When Cx43 hemichannels were studied in external solutions in which CO₂/bicarbonate was used to buffer pH instead of HEPES, Cx43 hemichannels showed significantly higher levels of dye uptake, ATP release, and ionic conductance. These findings may have major physiological implications since Cx43 hemichannels are found in many organs throughout the body, including the brain, heart, and immune system.

Weaknesses:

(1) Interpretation of the site-directed mutation studies is complicated. Although Cx43 has a potential carbamylation motif that is homologous to the motif in Cx26, the results of site-directed mutation studies were inconsistent with a simple model in which K144 and K105 interact following carbamylation to cause the opening of Cx43 hemichannels.

The mechanism of opening of Cx43 is more complex than that of Cx26, Cx32 and Cx50 and involves more Lys residues. The 4 Lys residues in Cx43 that are involved in opening the hemichannel have their equivalents in Cx26, but in Cx26 these additional residues seem to be involved in the closing of the GJC rather than opening of the hemichannel (see above). Cx50 is simpler and involves only two Lys residues (doi: 10.1101/2025.01.23.634273), which are equivalent to those in Cx26.

(2) Secondly, although it is shown that two Cx43 ODDD-associated mutations show a loss of CO₂ sensitivity, there is no evidence that the absence of CO₂ sensitivity is involved in the pathology of ODD

We agree, but this is probably because this has not been directly tested by experiment, as the CO₂ sensitivity of Cx43 was not previously known. As mentioned above we have revised the text to ensure that this is clear.

Reviewer #3 (Public review):

In this paper, the authors aimed to investigate carbamylation effects on the function of Cx43-based hemichannels. Such effects have previously been characterized for other connexins, e.g., for Cx26, which display increased hemichannel (HC) opening and closure of gap junction channels upon exposure to increased CO₂ partial pressure (accompanied by increased bicarbonate to keep pH constant).

The authors used HeLa cells transiently transfected with Cx43 to investigate CO₂ dependent carbamylation effects on Cx43 HC function. In contrast to Cx43-based gap junction channels that are reported here to be insensitive to PCO₂ alterations, they provide evidence that Cx43 HC opening is highly dependent on the PCO₂ pressure in the bath solution, over a range of 20 up to 70 mmHg encompassing the physiologically normal resting level of around 40 mmHg. They furthermore identified several Cx43 residues involved in Cx43 HC sensitivity to PCO₂: K105, K109, K144 & K234; mutation of 2 or more of these AAs is necessary to abolish CO₂ sensitivity. The subject is interesting and the results indicate that a fraction of HCs is open at a physiological 40 mmHg PCO₂, which differs from the situation under HEPES buffered solutions where HCs are mostly closed under resting conditions. The mechanism of HC opening with CO₂ gassing is linked to carbamylation, and the authors pinpointed several Lys residues involved in this process.

Overall, the work is interesting as it shows that Cx43 HCs have a significant open probability under resting conditions of physiological levels of CO₂ gassing, probably applicable to the brain, heart, and other Cx43 expressing organs. The paper gives a detailed account of various experiments performed (dye uptake, electrophysiology, ATP release to assess HC function) and results concluded from those. They further consider many candidate carbamylation sites by mutating them to negatively charged Glu residues. The paper ends with hippocampal slice work showing evidence for connexin-dependent increases of the EPSP amplitude that could be inhibited by HC inhibition with Gap26 (Figure 10). Another line of evidence comes from the Cx43-linked ODDD genetic disease, whereby L90V as well as the A44V mutations of Cx43 prevented the CO₂-induced hemichannel opening response (Figure 11). Although the paper is interesting, in its present state, it suffers from (i) a problematic Figure 3, precluding interpretation of the data shown, and (ii) the poor use of hemichannel inhibitors that are necessary to strengthen the evidence in the crucial experiment of Figure 2 and others.

The panels in Figure 3 were mislabelled in the accompanying legend possibly leading to some confusion. This has now been corrected.

We disagree that hemichannel blockers are needed to strengthen the evidence in Figure 2 and other figures. Our controls show that the CO₂-sensitive responses absolutely requires expression of Cx43 and was modified by mutations of Cx43. It is hard to see how this evidence would be strengthened by use of peptide inhibitors or other blockers of hemichannels that may not be completely selective.

Reviewing Editor Comments:

(1) Improve electrophysiological evidence, addressing concerns about the initial experiment and including peptide inhibitor data where applicable.

We think the concerns about the electrophysiological evidence arise from a misunderstanding because we gave insufficient information about how we conducted the experiments. We have now provided a much more complete legend, added explanations in the text and given more detail in the Methods. We further respond to the reviewer below.

We do not agree on the necessity of the peptide inhibitor to demonstrate dependence on Cx43. We have shown that parental HeLa cells do not release ATP to changes in PCO₂ or voltage (Fig 2D; Butler & Dale 2023, 10.3389/fncel.2023.1330983; Lovatt et al 2025, 10.1101/2025.03.12.642803, 10.1101/2025.01.23.634273). Our previous papers have shown many times that parental HeLa cells do not load with dye to CO₂ or zero Ca²⁺ (e.g. Huckstepp et al 2010, 10.1113/jphysiol.2010.192096; Meigh et al 2013, 10.7554/eLife.01213; Meigh et al 2014, 10.7554/eLife.04249), and we have shown that parental HeLa cells do not exhibit the same CO₂ dependent change in whole cell conductance that the Cx43-expressing cells do (Fig 2B). In addition, we shown that mutating key residues in Cx43 alters both CO₂-sensitive release of ATP and the CO₂-dependent dye loading without affecting the respective positive control. To bolster this, we have included data for the K144R mutation as a supplement to Fig 3. Given the expense of Gap26 it is impractical to include this as a standard control and unnecessary given the comprehensive controls outlined.

Collectively, these data show that the responses to CO₂ require expression of Cx43 and can be modified by mutation of Cx43.

(2) Strengthen the manuscript by measuring the effects of CO on cytosolic pH and Cx43 hemichannel opening. Consider using tail truncation mutants to assess the role of the C-terminal pH sensor in CO-mediated channel opening.

We agree and have performed the suggested experiments to address this issue.

(3) Investigate the effect of expressing the K105E/K109E Cx43 double mutant on cell viability.

In our experiments the cells look completely healthy based on their morphology in brightfield microscopy and growth rates.

(4) Discuss and analyze the uniqueness of Cx43 among alpha connexins in maintaining the carbamylation motif.

now discuss this -Cx43 is not unique. We have added a molecular phylogenetic survey of the alpha connexin clade in Fig 12. Apart from Cx37, the carbamylation motif appears in all the other members of the clade (but not necessarily in the human orthologue). In a different MS, currently posted on bioRxiv, we have documented the CO₂ sensitivity of Cx50 and its dependence on the motif.

(5) Consider omitting data on ODDD-associated mutations unless there is evidence linking CO₂ sensitivity to disease pathology.

This experiment is observational, and we are not making claims that there is a direct causal link. Removing the ODDD mutant findings would lose potentially useful information for anyone studying how these mutations alter channel function. We have reworded the text to ensure that we say that the link between loss of CO₂ sensitivity and ODDD remains unproven.

(6) Justify the choice of high K⁺ and low external calcium as a positive control in ATP release experiments.

These two manipulations can open the hemichannel independently of the CO₂ stimulus. Extracellular Ca²⁺ is well known to block all connexin hemichannels, and Cx43 is known to be voltage sensitive. The depolarisation from high K⁺ is effective at opening the hemichannel and we preferred this as a more physiological way of opening the Cx43 hemichannel. We have added some explanatory text.

(7) Clarify whether Cx43A44V or Cx43L90V mutations block gap junctional coupling.

This is an interesting point. Since Cx43 GJCs are not CO₂ sensitive we feel this is beyond the scope of our paper.

(8) Discuss the potential implications of pCO₂ changes on myocardial function through alterations in intracellular pH.

We have modified the discussion to consider this point.

Reviewer #1 (Recommendations for the authors):

(1) Measurements of the effects of CO₂ on cytosolic pH/Cx43 hemichannel opening would strengthen the manuscript. Since the pH sensor of Cx43 is on the C terminus, the authors could consider making tail truncation mutants to see how this affects CO₂-mediated Cx43 channel opening.

We have done this (truncating after residue 256) -the channel remains highly CO₂ and voltage sensitive. We have also documented the effect of the hypercapnic solutions on intracellular

pH measured with BCECF. These new data are now included as figure supplements to Figure 2.

(2) What is the impact of expressing the K105E / K109E Cx43 double mutant on cell viability?

There was no obvious observed impact, cell density was as expected (no evidence of increased cell death), brightfield and fluorescence visualisation indicated normal healthy cells. We have added a movie (Fig 9, movie supplement 1) to show the effect of La^{3+} on the GRAB_{ATP} signal in cells expressing Cx43^{K105E, K109E} so readers can appreciate the morphology and its stability during the recording.

(3) A quick look at other alpha connexins suggested that Cx43 was unique among alpha connexins in maintaining the carbamylation motif. This merits additional discussion/analysis.

This is an interesting point. Cx43 is not unique in the alpha clade in having the carbamylation motif as a number of other human alpha connexins also possess: Cx50, Cx59 and Cx62, and non-human alpha connexins (Cx40, Cx59, Cx46) also possess the motif. We have shown that Cx50 is CO₂ sensitive. We have performed a brief molecular phylogenetic analysis of the alpha connexon clade to highlight the occurrence of the carbamylation motif. This is now presented as Fig 12 to go with the accompanying discussion.

(4) There were some minor writing issues that should be addressed. For instance, fEPSP is not defined. Also, insets showing positive controls in some experiments were not described in the figure legends.

We have corrected these issues.

Reviewer #2 (Recommendations for the authors):

(1) I would omit the data on the ODDD-associated mutations since there is no evidence that loss of CO₂ sensitivity plays an important role in the underlying disease pathology.

We are not making the claim CO₂ loss leads to the underlying pathology and have reviewed the text to ensure that we clearly express that this is a correlation not a cause. We think this is worth retaining as many pathological mutations in other CO₂ sensitive connexins (Cx26, Cx32 and Cx50) cause loss of CO₂ sensitivity, and this information may be helpful to other researchers.

(2) Why is high K⁺ rather than low external calcium used as a positive control in ATP release experiments?

We used of high K⁺ and depolarisation as a positive control as regard this as a more physiological stimulus than the low external Ca²⁺.

(3) Does Cx43A44V or Cx43L90V block gap junctional coupling?

An interesting question but we have not examined this.

(4) Provide references for biophysical recordings of Cx43 hemichannels performed in HEPES-buffered salines, which document Cx43 hemichannels as being shut.

have added the original and some later references which examine Cx43 hemichannel gating in HEPES buffer and shows the need for substantial depolarisation to induce channel opening.

(5) In the heart muscle, changes in PCO_2 have long been hypothesized to cause changes in myocardial function by changing pH.

This is true and we now add some discussion of this point. Now that we know that Cx43 is directly sensitive to CO_2 a direct action of CO_2 cannot be ruled out and careful experimentation is required to test this possibility.

Reviewer #3 (Recommendations for the authors):

(1) Page 3: "... homologs of K125 and R104 ... ": the context is linked to Cx26, so Cx26 needs to be added here.

Done

(2) Page 4 text and related Figure 2:

(a) Figure 2A&B: PCO_2 -dependent Cx43 HC opening is clearly present in the carboxy-fluorescein dye uptake experiments (Figure 2A) as well as in the electrophysiological experiments (Figure 2B). The curves look quite different between these two distinct readouts: dye uptake doubles from 20 to 70 mmHg in Figure 2A while the electrophysiological data double from 45 to 70 mmHg in Figure 2B. These responses look quite distinct and may be linked to a non-linearity of the dye uptake assay or a problem in the electrophysiological measurements of Figure 2B discussed in the next point.

Different molecules/ions may have different permeabilities through the channel, which could explain the observed difference. Also, there is some contamination of the whole cell conductance change with another conductance (evident in recordings from parental HeLa cells). This is evident particularly at 70 mmHg. If this contaminating conductance were subtracted from the total conductance in the Cx43 expressing cells, then the dose response relations would be more similar. However, we are reluctant to add this additional data processing step to the paper.

(b) The traces in Figure 2B show that the HC current is inward at 20 mmHg PCO_2 , while it switches to an outward current at 55mmHg PCO_2 . HCs are non-selective channels, so their current should switch direction around 0 mV but not at -50 mV. As such, the -50 mV switching point indicates involvement of another channel distinct from non-selective Cx43 hemichannels.

We think that our incomplete description in the legend led to this misunderstanding. We used a baseline of 35 mmHg (where the channels will be slightly open) and changed to 20 mmHg to close them (or to higher PCO_2 to open them from this baseline), hence a decrease in conductance and loss of outward current for 20 mmHg. The holding potential for the recordings and voltage steps were the same in all recordings. We have now edited the legend and added more information into the methods to clarify this and how we constructed the dose response curve.

We agree that Cx43 hemichannels are relatively nonselective and would normally be expected to have a reversal potential around 0 mV, but we are using K-Gluconate and the lowered reversal potential (~-65 mV) is likely due to poor permeation of this anion via Cx43.

(c) A Hill slope of 6 is reported for this curve, which is extremely steep. The paper does not provide any further consideration, making this an isolated statement without any theoretical framework to understand the present finding in such context (i.e., in relation to the PCO₂ dependency of Cx channels).

Yes, we agree -it seems to be the case with all CO₂ sensitive connexins that we have looked at that the Hill coefficient versus CO₂ is >4. Hemichannels are of course hexameric so there is potential for 6 CO₂ molecules to be bound and extensive cooperativity. We have modified the text to give greater context.

(d) A further remark to Figure 2 is that it does not contain any experiment showing the effect of Cx43 hemichannel inhibition with a reliable HC inhibitor such as Gap26, which is only used in the penultimate illustration of Figure 10. Gap26 should be used in Figure 2 and most of the other figures to show evidence of HC contribution. The lanthanum ions used in Figure 9 are a very non-specific hemichannel blocker and should be replaced by experiments with Gap26.

We have addressed the first part of this comment above.

We agree that La³⁺ blocks all hemichannels, but in the context of our experiments and the controls we have performed it is entirely adequate and supports our conclusions. Our controls show (mentioned above and below) show that the expression of Cx43 is absolutely required for CO₂-dependent ATP release (and dye loading). In Figure 9 our use of La³⁺ was to show the presence of a constitutively open Cx43 mutant hemichannel. Gap26 would add little to this. Our further controls show that with expression of Cx43^{WT} La³⁺ did nothing to the ATP signal under baseline conditions (20 mmHg) supporting our conclusion that the mutant channels are constitutively open.

(e) As the experiments of Figure 2 form the basis of what is to follow, the above remarks cast doubt on the robustness of the experiments and the data produced.

We disagree, our results are extremely robust: 1) we have used three independent assays confirm the presence of the response; 2) parental HeLa cells do not release ATP, dye load or show large conductance changes to CO₂ showing the absolute requirement for expression of Cx43; 3) mutations of Cx43 (in the carbamylation motif) alter the CO₂ evoked ATP release and dye loading giving further confirmation of Cx43 as the conduit for ATP release and dye loading; and 4) we use standard positive controls (0 Ca²⁺, high K) to confirm cells still have functional channels for those mutations that modified CO₂ sensitivity.

(f) The sentence "Cells transfected with GRAB-ATP only, showed ..." should be

modified to "In contrast, cells not expressing Cx43 showed no responses to any applied CO₂ concentration as concluded from GRAB-ATP experiments"

We have modified the text.

(3) Page 5 and Figures 3 & 4:

(a) Figure 3 illustrates results obtained with mutations of 4 distinct Lys residues. However, the corresponding legend indicates mutations that are different from the ones shown in the corresponding illustrations, making it impossible to reliably understand and interpret the results shown in panels A-E.

Thanks for pointing this out. Our apologies, we modified the figure so that the order of the images matched the order of the graph (and the legend) but then forgot to put the new version of the figure in the text. We have now corrected this so that Figure and legend match.

(b) Figure 4 lacks control WT traces!

The controls for this (showing that parental HeLa cells do not release ATP in response to CO₂ or depolarisation) are shown in Figure 2.

(c) Figure 4, Supplement 1: High Hill coefficients of 10 are shown here, but they are not discussed anywhere, as is also the case for the remark on p.4. A Hill steepness of 10 is huge and points to many processes potentially involved. As reported above, these data are floating around in the manuscript without any connection.

Yes, we agree this is very high and surprising. It may reflect as mentioned above the hexameric nature of the channel and that 4 Lys residues seem to be involved. We have used this equation to give some quantitative understanding of the effect of the mutations on CO₂ sensitivity and still think this is useful. We have no further evidence to interpret these values one way or the other.

(4) Page 6: Carbamate bridges are proposed to be formed between K105 and K144, and between K109 and K234. The first three of these Lysine residues are located in the 55aa long cytoplasmic loop of Cx43, while K234 is in the juxta membrane region involved in tubulin interactions. Both K144 and K234 are involved in Cx43 HC inhibition: K144 is the last aa of the L2 peptide (D119-K144 sequence) that inhibits Cx43 hemichannels while K234 is the first aa of the TM2 peptide that reduces hemichannel presence in the membrane (sequence just after TM4, at the start of the C-tail). This context should be added to increase insight and understanding of the CO₂ carbamylation effects on Cx43 hemichannel opening.

Thanks for suggesting this. We have added some discussion of CT to CL interactions in the context of regulation by pH and [Ca²⁺].

(5) Page 7: The Cx43 ODDD A44V and L90V mutations lead to loss of pCO₂ sensitivity in dye loading and ATP assays. However, A44V located in EL1 is reportedly associated with Cx43 HC activation, while L90V in TM2 is associated with HC inhibition. Remarkably, these mutations are focused on non-Lys residues, which brings up the question of how to link this to the paper's main thread.

This follows the pattern that we have seen for other mutations such as A40V, A88V in Cx26 and several CMTX mutations of Cx32. Our cryoEM structures of Cx26 suggest that these mutations alter the flexibility of the molecule and hence abolish CO₂ sensitivity. We have reworded the text to avoid giving the impression that there is a demonstrated link between loss of CO₂ sensitivity of Cx43 and pathology.

(6) Page 8: HCs constitutively open - 'constitutively' perhaps does not have the best connotation as it is not related to HC constitution but CO₂ partial pressure.

Yes, we agree and have reworded this.

(7) Page 9: "in all subtypes" -> not clear what is meant - do you mean "in all cell types"?

We agree this is unclear -it refers to all astrocytic subtypes. We have amended the text.

(8) Page 10: Composition of hypocapnic recording solution: bubbling description is incomplete "95%O₂/5%" and should be "95%O₂/5%CO₂".

Changed.

(9) Page 11: Composition of zero Ca²⁺ hypocapnic recording solution: perhaps better to call this "nominally Ca²⁺-free hypocapnic recording solution" as no Ca²⁺ buffer is included in this solution

Thanks for pointing this out. We did in fact add 1 mM EGTA to the solutions but omitted this from the recipe, this has now been corrected.

(10) Page 11: in M&M I found that the NaHCO₃- is lowered to 10 mM in the zero Ca²⁺ condition, while the control experimental condition has 26 mM NaHCO₃-. The zero Ca condition should be kept at a physiologically normal 26 mM NaHCO₃- concentration, so why was this done? Lowering NaHCO₃- during hemichannel stimulation may result in smaller responses and introduce non-linearities.

For the dye loading we used 20 mmHg as the baseline condition and increased PCO₂ from this. Hence for the zero Ca²⁺ positive control we modified the 20 mmHg hypocapnic solution by substituting Mg²⁺ for Ca²⁺ and adding EGTA. We have modified the text in the Methods to clarify this.

Further remarks on the figures:

(1) Figure 2A: Add 20 & 70 mmHg to the images, to improve the readability of this illustration.

Done

(2) Figure 3: WT responses are shown in panel F, but experimental data (images and curves) are lacking and should be included in a revised version.

The wild type data is shown in Fig 2A. We have some sympathy for the comment, but we felt that Fig 2 should document CO₂ sensitivity, and then the subsequent Figs should analyse its basis. Hence the separation of Cx43^{WT} data from the mutant data. In panel F, we state that we have recalculated the WT data from Fig 2A to allow the comparison.

(3) Figures 4, 6, 8: Color codes for mmHg CO₂ pressure make reading these figures difficult; perhaps better to add mmHg values directly in relation to the traces.

We have considered this suggestion but feel that the figures would become very cluttered with the additional labelling.

(4) I wouldn't use colored lines when not necessary, e.g., Figure 9 100 μM La3+; Figure 10 (add 20->35 mmHg PCO₂ switch; add scrGap26 above blue bars); Figure 11C & D.

We agree and can see that in Figs 9 and 10 this muddles our colour scheme in other figures so have modified these figures. There was not space to put the suggested labels.

(5) The mechanism of increased HC opening is not clear.

We agree and have discussed various options and the analogy with what we know about Cx26. Ultimately new cryo-EM data is required.

| (6) Figure 10: 35G/35S are weird abbreviations for 35 mmHg Gap26 and scrambled Gap26.

Yes, but we used these to fit into the available space.

| (7) Figure 11, legend: '20 mmHg PCO₂ for each transfection for 70 mmHg PCO₂'. It is not clear what is meant here.

Thanks for pointing this out, we have reworded this to ensure clarity.

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